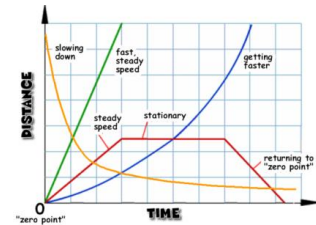


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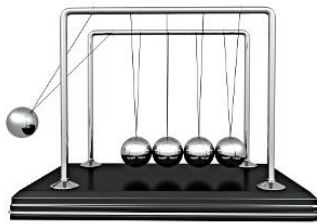
Form:



Physics



FORCES



DOUBLE AWARD

FORCE AND MOTION



Push



Pull



Magnetism



Gravity



Friction



Acceleration

2026 onward

Lesson 1: Units and Vectors

Lesson 2: Speed

Lesson 3: Acceleration

Lesson 4: Motion Graphs

Lesson 5: Acceleration Experiment

Lesson 6: Acceleration Calculation: SUVAT

Lesson 7: Terminal Velocity

Lesson 8: Stopping Distance

Lesson 9 and 10: Hooke's Law



Forces – Double Award

Ref	Learning Objective	C	P	N
1.1	I can use the following units : Kilogram (kg), metre (m), metre/second (m/s), Newton (N), second (s), and Newton/kilogram (N/kg)			
1.3	I can plot and explain distance-time graphs.			
1.4	I know and can use the equation Average speed = distance travelled / time taken			
1.5	<i>Practical: I can investigate the motion of everyday objects such as toy cars and tennis balls</i>			
1.6	I know and can use the equation acceleration = change in velocity / time or a = (v – u) / t			
1.7	I can plot and explain velocity-time graphs Page.....			
1.8	I can determine acceleration from the gradient of a velocity-time graph			
1.9	I can determine distance travelled from the area under a velocity-time graph			
1.10	I can use the relationship: $(\text{final speed})^2 = (\text{initial speed})^2 + (2 \times \text{acceleration} \times \text{distance})$ or $v^2 = u^2 + 2.a.s$			
1.11	I can describe the effect that forces have on objects, eg forces cause changes in speed, changes of shapes or changes of direction.			
1.12	I can identify types of force such as gravitational, electrostatic.			
1.13	I understand how Vector quantities differ from Scalar quantities.			
1.14	I understand that force is a vector .			
1.15	I can calculate the Resultant Force of forces along a line.			
1.16	I know that Friction is a force that opposes motion .			
1.17	I know and can use the formula Force = mass x acceleration or F = m x a (Newton's Second Law)			
1.18	I know and can use the formula Weight = mass x gravitational field strength or W = m x g			
1.19	I know that stopping distance = thinking distance + braking distance			
1.20	I can describe the factors affecting stopping distance including speed , mass , road condition and reaction time			
1.21	I can describe the forces acting on falling objects and explain why they reach terminal velocity			
1.22	<i>Practical: I can investigate how extension varies with applied force for helical springs, metal wires and rubber bands.</i>			
1.23	I know that the initial linear region of a force-extension graph is associated with Hooke's Law .			
1.24	I can describe elastic behaviour as the ability of a material to recover its original shape once a deforming force is removed			
Self Assessment Score yourself. C = 2, P = 1, N = 0				%
Targets Set three targets to help you improve in this topic: • • •				

Forces : Lesson 1 :Units and Vectors

Learning Objectives:

1. I know the lab safety rules
2. I know how to access the IGCSE course information and other resources on [ISLE](#)
3. I have a textbook and I am ready to rock and roll...
4. I know the symbols and units for the key quantities in this topic
5. I know the formula for speed and acceleration
6. I know the difference between scalars and vectors
7. I can add vectors

Key quantities: distance s (m), time t (s), speed v (m/s), acceleration a (m/s²), force F (N), mass m (kg),

1.1	I can use the following units : Kilogram (kg), metre (m), metre/second (m/s), Newton (N), second (s), and Newton/kilogram (N/kg)
1.11	I can describe the effect that forces have on objects, eg forces cause changes in speed, changes of shapes or changes of direction. Speed = distance / time ($v = d/t$), acceleration = change in velocity ÷ time ($a = (v - u)/t$).
1.12	I can identify types of force such as gravitational, electrostatic, gravitational (weight), electrostatic, magnetic, friction, air resistance (drag), tension, upthrust.
1.14	I understand that force is a vector . Force is a vector - it has size and direction, so 5 N to the left is not the same as 5 N to the right.
1.15	I can calculate the Resultant Force of forces along a line. Same direction: add ($3\text{ N} + 4\text{ N} = 7\text{ N}$). Opposite: subtract ($7\text{ N} - 4\text{ N} = 3\text{ N}$, acting in the direction of the larger force).

Page reference: P11-13 Hodder Book,

Units

Quantity	Symbol	Unit	Unit Symbol
Mass	m	Kilograms	kg
Length, distance, displacement	Symbol: s (or d) Unit symbol: m	Metres	
Time	Symbol: t Unit symbol: s	Second	
Force	Symbol: F Unit symbol: N	Newton	
Speed / Velocity	Symbol: v Unit symbol: m/s	Metre per second	
Acceleration	Symbol: a Unit symbol: m/s^2	Metres per second squared	

Add vectors tip-to-tail: the resultant runs from the start of the first to the end of the last. Along a line, add forces in the same direction and subtract opposite ones.

Forces change an object's speed, direction of motion or shape - they can start, stop, speed up, slow down or turn an object.

Speed = distance / time

Acceleration =

Scalars and Vectors

Some properties have only **size**..... They are called **SCALARS**.

Some properties have both **Size and direction**..... They are **VECTORS**.

Where would you place: Mass Time Length, distance Speed Displacement Velocity
Acceleration Force

<u>Scalar</u> Scalar column: mass, time, length, distance, speed	<u>Vector</u> Vector column: displacement, velocity, acceleration, force
mass	

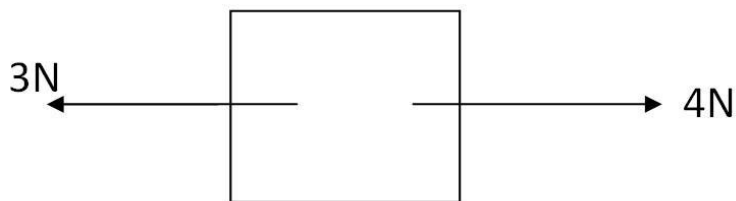
Adding Scalars and Vectors.

Adding scalars is easy.....

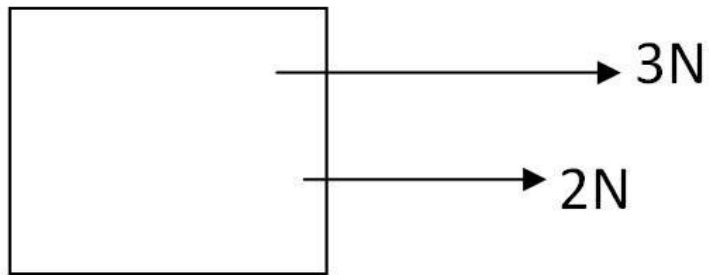
What is 2kg of apples plus 5kg of apples? 7..... kg of apples

Adding Forces (this works for any vectors)

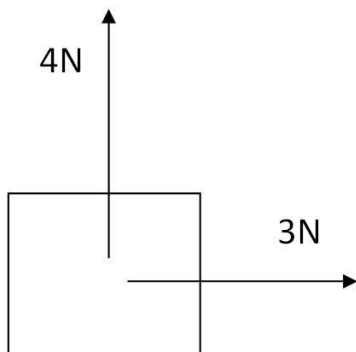
What is 3N add 4N? 7 N if the two forces act in the same direction; 1 N if they act in opposite directions (4 N - 3 N).



What is 2N add 3N? 5 N if the forces act in the same direction; 1 N in the direction of the 3 N force if they act in opposite directions.



What is 3N add 4N ? Draw them tip-to-tail to scale: 3 N and 4 N at right angles give a resultant of 5 N; if in line, 7 N (same direction) or 1 N (opposite).

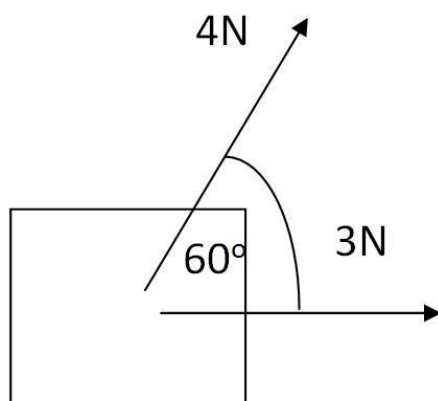


How do we add vectors in 2 dimensions? (not required for IGCSE)

The 5 easy pieces:

- Choose a scale
- Choose a start point
- Draw the first vector
- **ADD ON** the second vector
- Draw and measure the resultant from **start to finish!**

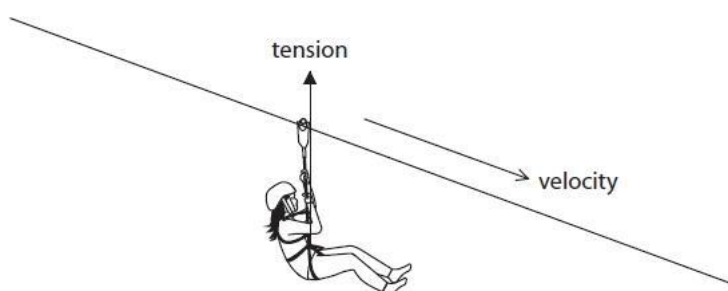
Extension - Apply the same technique to this question:



Practice Question:

The diagram shows a child on a zip-line ride.

During the ride the child slides along a metal wire with the velocity in the direction shown in the diagram.



The diagram also shows the tension force acting on the safety harness the child is wearing.

(a) Force and velocity are examples of vector quantities.

State what is meant by the term **vector quantity**.

(1)

~~A vector quantity has both size (magnitude) and direction.~~.....

(b) Draw labelled arrows on the diagram to show two other forces that act on the child.

(4)

(Total for question = 5 marks)

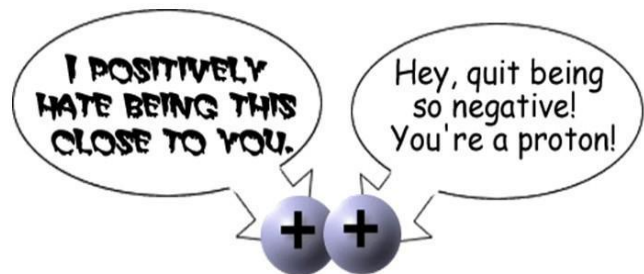
· Vector addition by scale drawing: draw each vector tip-to-tail at the angle given, then draw the resultant from the start of the first to the end of the second and measure its length (convert using the scale).

Forces and Fields

To apply a force to an object it can be pushed or pulled. However, sometimes a **non-contact force** can occur. This happens in **Fields (called Fields of Force)**.

In an **Electric Field** a **non-contact force**

occurs on a **charged object**..



ISAAC NEWTON DISCOVERS THE LAW OF GRAVITY



In an **Gravitational Field** a

non-contact force occurs on a **mass**.....

Homework; organize your file with index dividers

New Hodder Book IGCSE Physics P14 Q1-6

Optional:

Old Oxford IGCSE Physics Book P47, Q 1-2

Extension; P47, Q3

Remember Homework is green!

Forces: Lesson 2 : Speed

Learning Objectives:

- Recall and manipulate the formula for speed.
- Recall units for speed
- Develop practical skills including using a stopwatch and team skills
- Introduce the idea of uncertainties in our measurements.

1.4	I know and can use the equation $\text{average speed} = \text{distance} \div \text{time}$, $v = s/t$ (m/s); rearrange to find distance or time. Average speed = distance travelled / time taken
1.5	Practical: I can investigate the motion of everyday objects such as toy cars and tennis balls

Page ref Hodder Book P. 2-4

Speed Definition and Formula

Speed is the **DISTANCE** travelled in a certain **TIME**

$$\text{Speed} = \text{dist} / \text{time} \quad (\text{Can you rearrange this?})$$

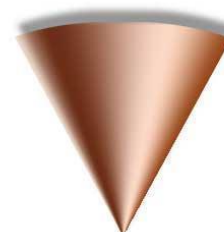
Units for speed: **m/s** or km/hr or m.p.h. or Mach
or knots or light speed

Speed Experiment - 1

Research Question: What is the speed of a falling paper cone?

Method:

1. Cut out a paper circle.
2. Cut a radius, fold into a cone and sellotape or glue in place.
3. Measure a known **height** from, say, the water pipes to the floor.
4. Measure the **time** of fall to the ground
5. **Repeat** an appropriate number of times.
6. Calculate an **average time** and hence **calculate speed**.
7. Discuss the uncertainties.



distance = speed x time time = distance / speed

Drop the cone from a measured height, time the fall with a stopwatch and repeat;
speed = height ÷ average time (e.g. 2.0 m ÷ 1.3 s = 1.5 m/s).

Results and Calculations:

Measured Height = 1.50 m

Please put your results in a table and work out an average.

Time, repeat 1 /s	Time, repeat 2 /s	Time, repeat 3 /s	Average time /s

Speed = distance/time

(use distance fallen and the average time to work out the speed of your cone in m/s)

Speed = 2.7 m/s (speed = distance / time = 1.50 m / 0.55 s)

Improving our Experiments

At IGCSE we need to know how to improve our experiments. There are two aspects to be aware of:

- **Accuracy.** This means how close our value is to the 'real' or 'accepted' value.
- **Reliability.** This is the confidence we can have in our reading

How can we improve accuracy and reliability?

Take repeat readings and average them, use more sensitive instruments, avoid zero errors, plot a graph with a line of best fit and discard anomalous results.

<u>Improving Accuracy</u>	<u>Improving Reliability</u>
This requires: • better experiment techniques, • avoiding zero errors • using more sensitive instruments....	This requires: • Repeat... readings, • Calculate an average..... • plotting results on graph..... with a line of best fit • Eliminating anomalies....
Suggestions:	Suggestions:

Accuracy: use better measuring instruments (e.g. light gates), avoid zero errors and parallax, and use a longer fall distance so timing errors matter less.

Reliability: take at least three repeat readings, calculate an average and ignore any anomalous results.

Speed Experiment – 2

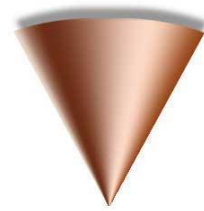
Research Question: What is the height of the 3rd floor at IS?

Method:

1. Measure the speed of fall of a paper cone (done)
2. Measure the time of fall from the 3rd floor to the ground floor
3. Repeat an appropriate number of times.
4. Calculate an average time and hence calculate height.
5. Discuss the uncertainties.

Average time =

Height = speed x average time = Height = speed x average time = 2.7 m/s x 3.2 s = 8.6 m (using the cone speed from Experiment 1)



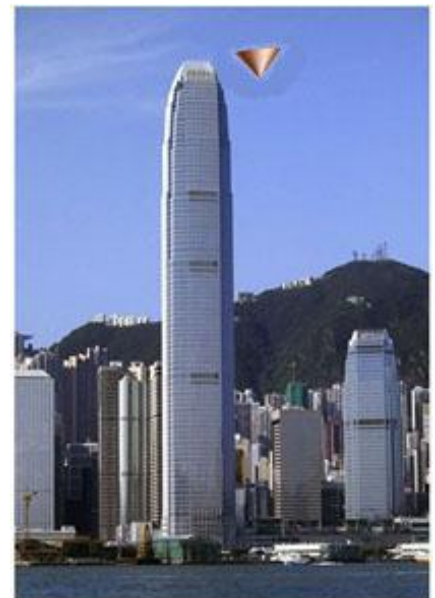
Experiment 3 – Results and Calculation

Speed Calculation - 3

Question: What would be the time of fall for a paper cone dropped off the 2-IFC (you can also choose some other buildings around the world if you like)?

Information:

Extension: use wikipedia drop link in the table to estimate other heights (Compare with famous building - This (20415%2D300m)%2Dtall%20(,extremely%20auspicious%20in%20Cantonese%20culture.



Formula:

Time = Time = distance / speed = 415 m / 2.7 m/s = 154 s (using the measured cone speed)

Batman has a terminal velocity of 53m/s. How long would it take Batman to reach the ground?

Time the cone falling from the 3rd floor several times, then height = speed of the cone x average time of fall.

Time = Time = distance / speed = 415 m / 53 m/s = 7.8 s

What is the airspeed velocity of an unladen swallow?

<http://www.youtube.com/watch?v=PW8Mg-wSg8c&feature=related>

BBC Bitesize: <https://www.bbc.co.uk/bitesize/guides/z3whb82/revision/1>

Homework:

Finish these q's. In the new Hodder Book: P 4 Q1-7.

Optional

Old Oxford Book P 27 Q 1-4. Extension: P 50 Q 1-2

Practice Question

A driver of a car sees an obstruction in the road ahead and must stop the car.

(a) (i) State the formula linking average speed, distance travelled and time taken.

(1)

(ii) A car travels at 21 m / s.

The driver's reaction time is 0.14 seconds.

Calculate the distance travelled by the car during the driver's reaction time.

(2)

distance = 2.94..... m

average speed = distance travelled / time taken

Forces : Lesson 3: Acceleration

Ticker Tape Experiment

Learning Objectives

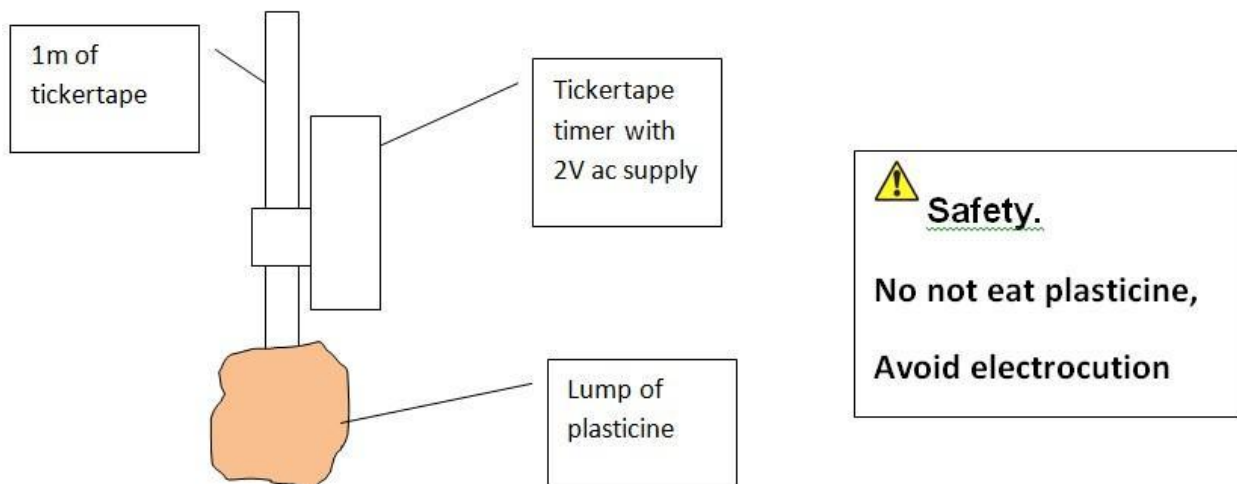
- Recall and use the formula for acceleration
- Learn the use of ticker tape for measuring speed and acceleration
- Find a value for the acceleration of free fall on earth

1.6	I know and can use the equation $\text{acceleration} = \text{change in velocity} \div \text{time}$, $a = (v - u)/t$ (m/s ²); equivalent to $v = u + at$. acceleration = change in velocity / time or $a = (v - u) / t$
-----	---

Page ref New Hodder Book P 8-10. But better spread on P 30-31 in Old blue Oxford Book if access is available.

Research question

What is the **acceleration due to gravity** as calculated by using the ticker tape timer?



Method

1. Set the powerpack to 2V a.c.
2. Allow the plasticine to fall whilst dragging ticker tape through the timer.
3. Choose two places along the ticker tape that are 10 dots apart.
4. Calculate the speed at start by measuring the distance between two consecutive dots.
5. Calculate the speed at end by measuring the distance between two consecutive dots
6. Calculate the **acceleration of gravity**.

Have a look at this youtube video to get an idea:

<https://www.youtube.com/watch?v=FcvW7PxqgTo>

Also look at book page 30 to 31 and answer the questions. (No 1)

Calculations

The speed calculation is quite easy:

$$\text{Speed} = \text{Distance} / \text{time}$$

The unit of speed is m/s

The timer makes 50 dots in one second. Time between dots is $1/50 = 0.02\text{s}$

The acceleration calculation is quite easy:

$$\text{Acceleration} = \text{change in } \text{velocity} / \text{time}$$

The unit of acceleration is m/s^2

(No 2)

Example:

$$d_0 = 1.2 \dots \text{cm} \quad \text{Initial speed} = \text{dist} / \text{time} = 0.012 \dots / 0.02 = 0.6 \dots \text{m/s}$$

$$d_{10} = 5.12 \dots \text{cm} \quad \text{Final speed} = \text{dist} / \text{time} = 0.0512 \dots / 0.02 \dots = 2.56 \dots \text{m/s}$$

$$\text{Acceleration} = (V - U) / t = (2.56 \dots - 0.6 \dots) / 0.2 = 9.8 \dots \text{m/s}^2$$

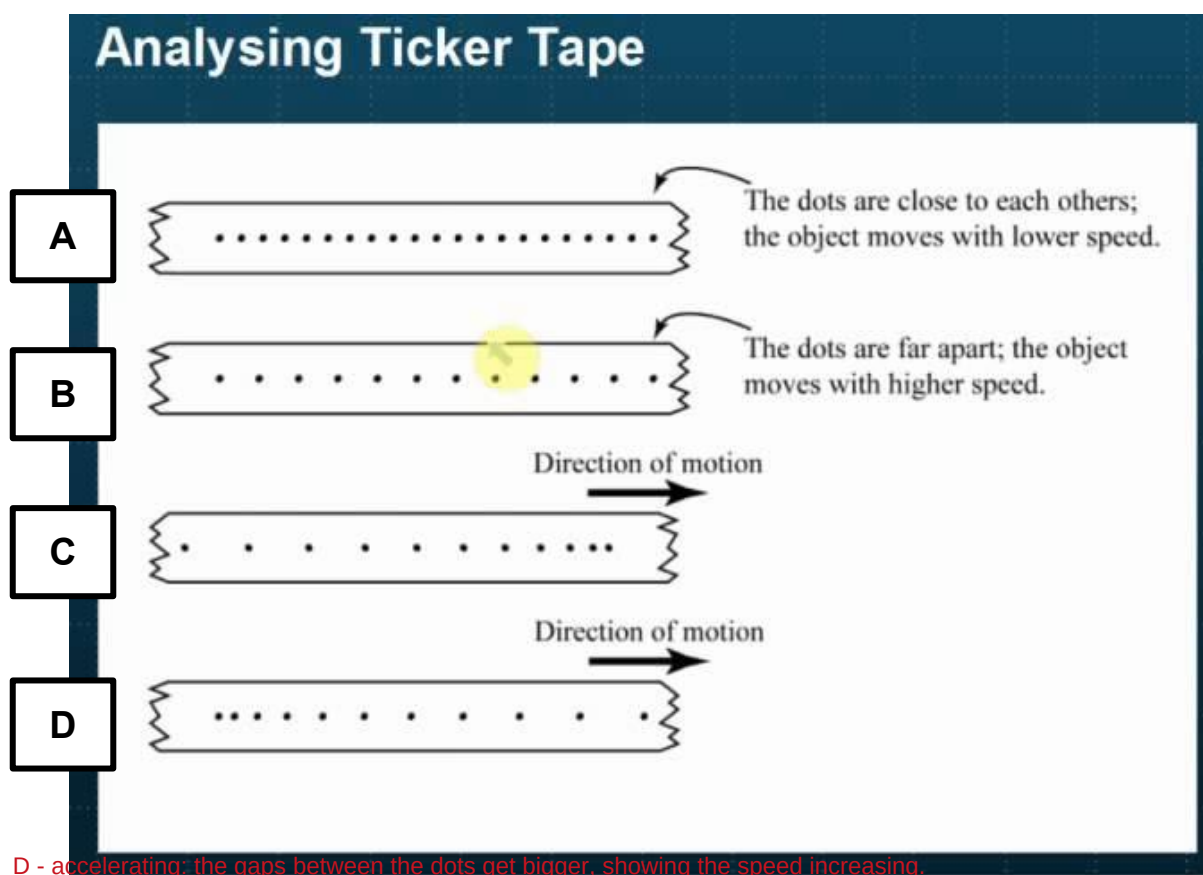
The accepted value for the acceleration due to gravity is 9.8 m/s^2 account for the difference in your value:

Friction between the tape and the timer, plus air resistance, slow the fall so the measured acceleration is slightly less than the accepted value.

Write answers here:

Ticker Tape Questions

Q1. Describe the motion of these objects:



1a. Explain which objects are moving at a constant speed?

1b. Which objects are showing an acceleration or a deceleration? Explain your answer.

A is at a constant speed (and any strip with equally spaced dots), because the distance travelled in each 0.02 s is the same.

B and D are accelerating (dots get further apart) and C is decelerating (dots get closer), because the distance travelled in equal times changes.

A - constant speed: the dots are equally spaced, so the same distance is travelled in the same time.

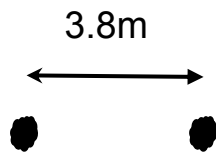
B - accelerating: the distance between the dots increases, so the speed is increasing.

C - decelerating: the distance between the dots decreases, so the speed is decreasing.

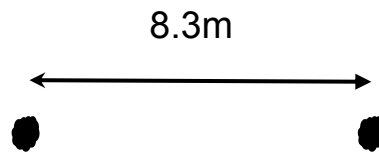
Q2. A car drives down the road. It is dripping oil from the engine, there is an oil drop every 1.4s.

The following measurements are taken:

Drops fall at equal time intervals, so equal gaps mean constant speed; gaps that get bigger mean the car is accelerating.



First 2 oils
drops



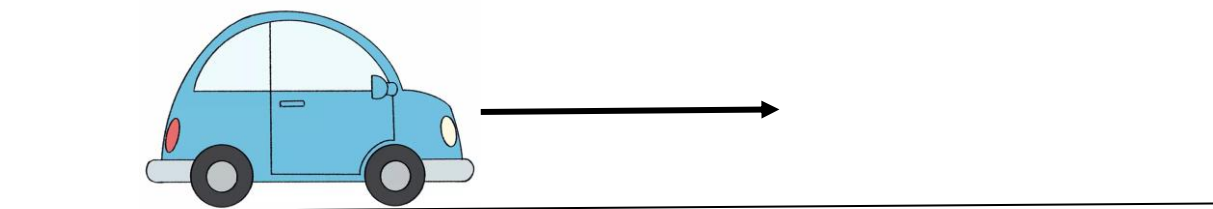
Oils drops measured
4.7s later

Calculate the acceleration of the car (3).

Practice Question:

A car drives down the road at 12m/s. It accelerates to 24m/s over 15 seconds.

Calculate its acceleration. $a = (v - u) / t = (24 - 12) / 15 = 0.8 \text{ m/s}^2$

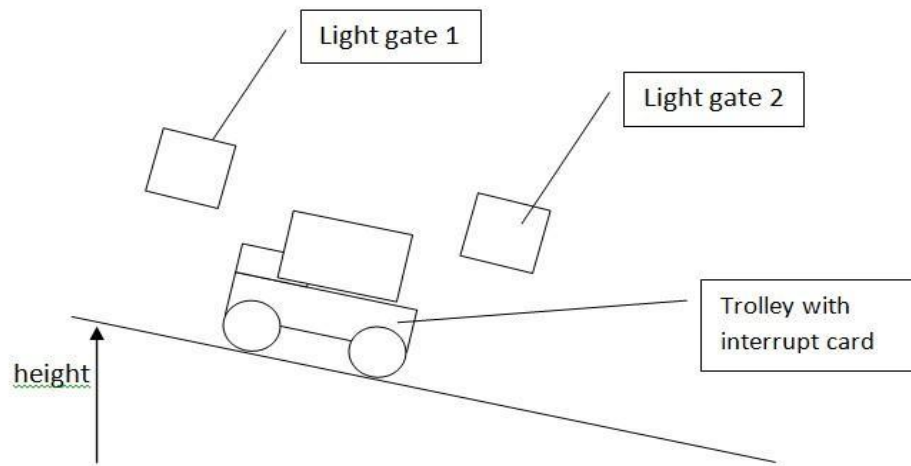


Speed 1 = $3.8 / 1.4 = 2.7 \text{ m/s}$; speed 2 = $8.3 / 1.4 = 5.9 \text{ m/s}$; $a = (v - u) / t = (5.9 - 2.7) / 4.7 = 0.68 \text{ m/s}^2$.

Light Gates Demonstration

What is the relationship between the height of a ramp and the acceleration of a trolley down the ramp?

Apparatus



Method

1. Set the ramp to a chosen height and ensure the light gates are at a correct height.
2. Allow the trolley to accelerate down the ramp.
3. Record data from the computer for the speed at the first and second light gate and also the time interval.
4. Calculate the acceleration.
5. Repeat the process and gain an average acceleration.
6. Continue for seven different heights of the ramp.
7. Plot a graph of Height (x-axis) against Acceleration (y-axis).

We can't do this but here is a video that shows how light gates work:

<https://www.youtube.com/watch?v=xdKufK65byM>

Now and for Homework:

Finish the graph

P9 Q1 & 2 in New Hodder Book

P31 Q1-3 in Old Blue Oxford Book

The greater the height of the ramp, the greater the acceleration of the trolley - acceleration is directly proportional to the height of the ramp.

Forces : Lesson 4 : Motion Graphs

Learning Objectives

- I can use **Distance/time graphs**
- I know the significance of the gradient on a distance/time graph
- I can use **velocity/time graphs**
- I know the significance of the gradient on a velocity/time graph
- I know the significance of the area under a velocity/time graph

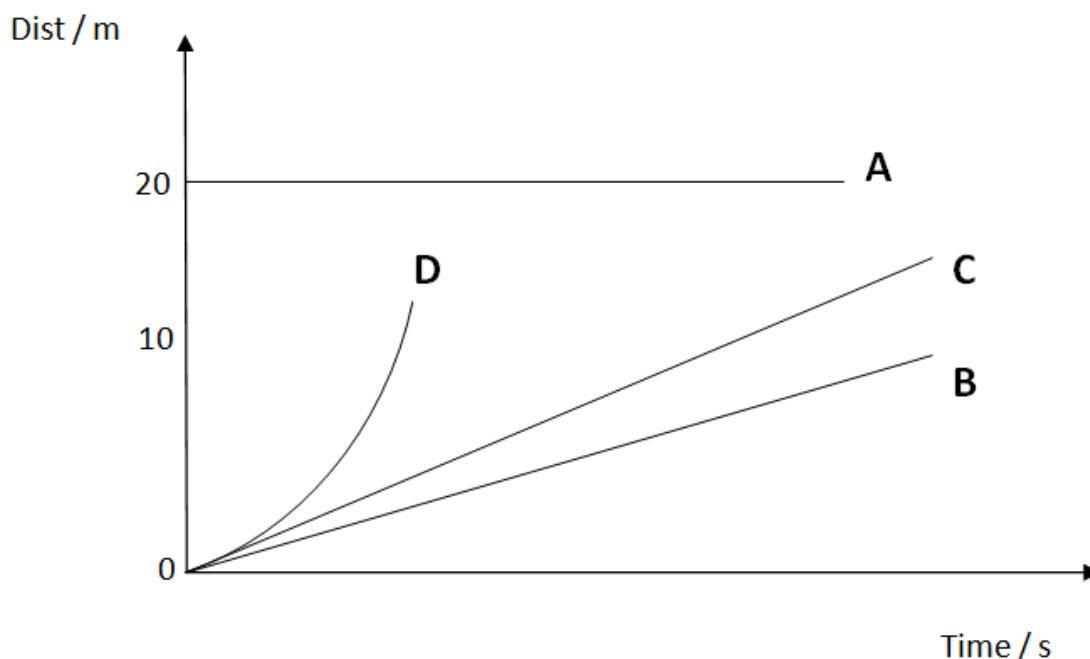
1.7	I can plot and explain velocity-time graphs
1.8	I can determine acceleration from the gradient of a velocity-time graph
1.9	I can determine distance travelled from the area under a velocity-time graph

Page ref 2-6 New Hodder Book

Types of Motion Graphs There are two types of motion graphs:

- ❖ Distance / time graphs also called displacement / time graphs
- ❖ Speed / time graphs also called velocity / time graphs

Distance / Time Graphs



Object A is **stationary**..... because the line is **horizontal**.....

Object B is moving away at a constant **speed**..... because it is a **straight**..... line going up

Velocity-time graph: velocity on the y-axis, time on the x-axis.

Horizontal line = constant velocity; straight slope = uniform acceleration; curve = changing acceleration.

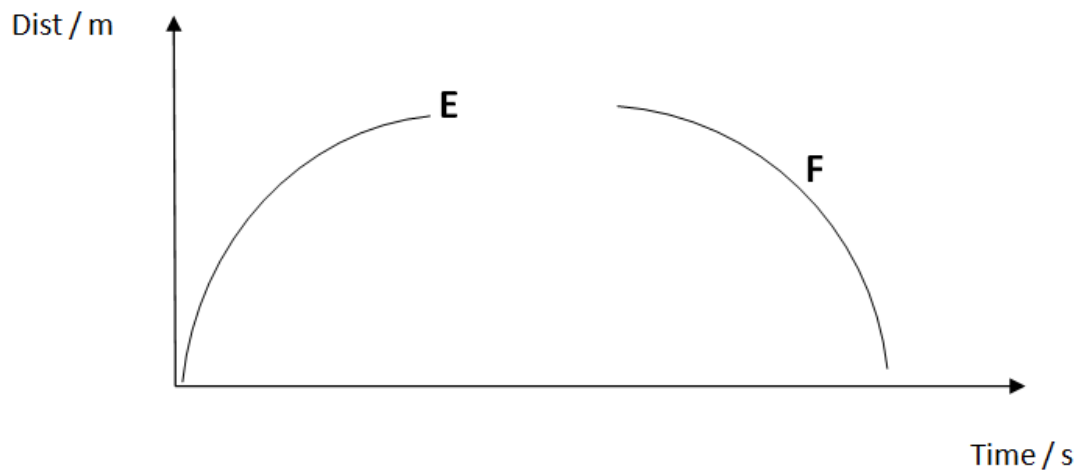
Acceleration = gradient of the velocity-time graph = change in velocity ÷ change in time.

Distance travelled = area under the velocity-time graph (split it into triangles and rectangles and add them).

Object C is also moving at a constant speed..... which is faster... than the velocity of object B, because it is also a straight..... line, which is steeper..... than B.

Object D is accelerating..... because the line is becoming steeper..... and steeper.....

as time goes on.



What's happening with E and F?

E is stationary

F isF is moving back towards the start - the line slopes downwards, so the distance from the start is decreasing.

Conclusion

On a distance time graph the gradient..... is the velocity

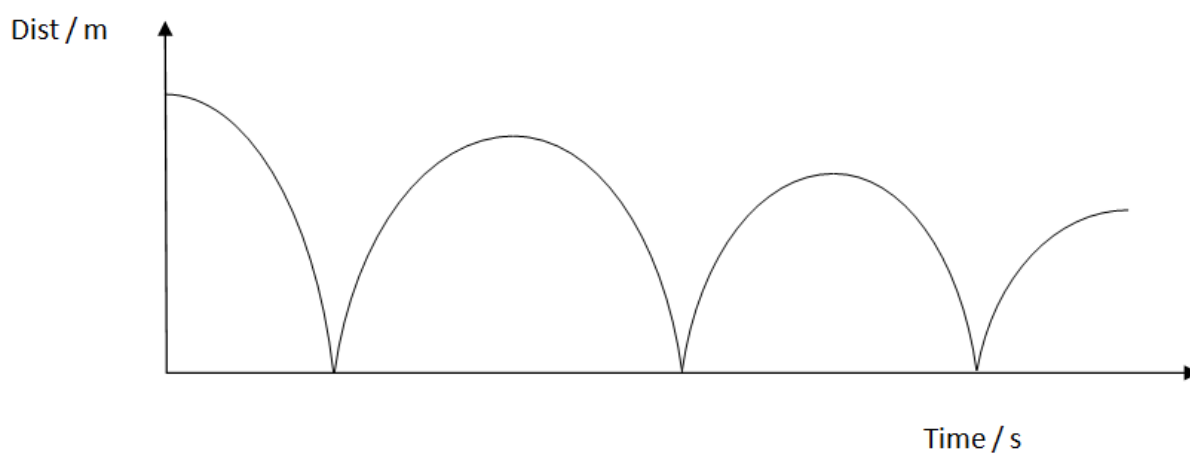
Extension-(difficult)

What is a dist/time graph for a bouncing ball? Three bounces, ball is released from 1m above ground.

Forces 1.4 (Double Award)

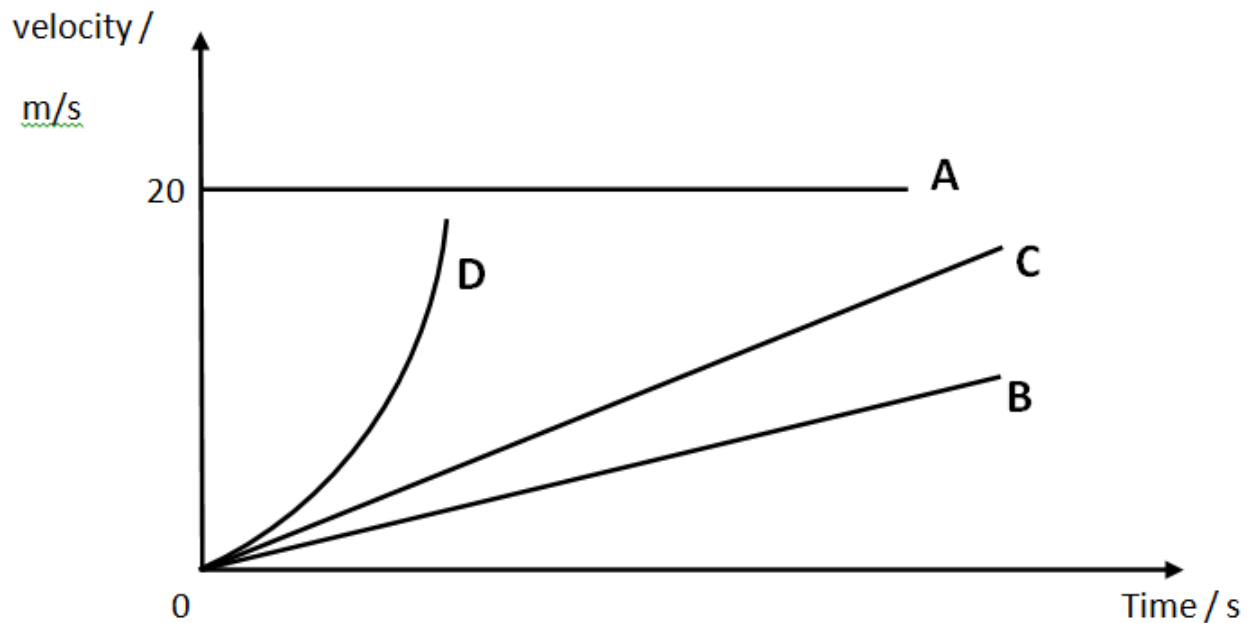


Answer for bouncing ball:



· Distance-time graph for a bouncing ball: starts at 1 m, falls as a steep curve flattening at the top of each bounce; each bounce rises to a lower height than the last (three decreasing peaks) because energy is lost on each impact.

Velocity / Time Graphs



Object A is moving at a **constant..... velocity.....** of, 20 m/s, because the line is **horizontal.....**

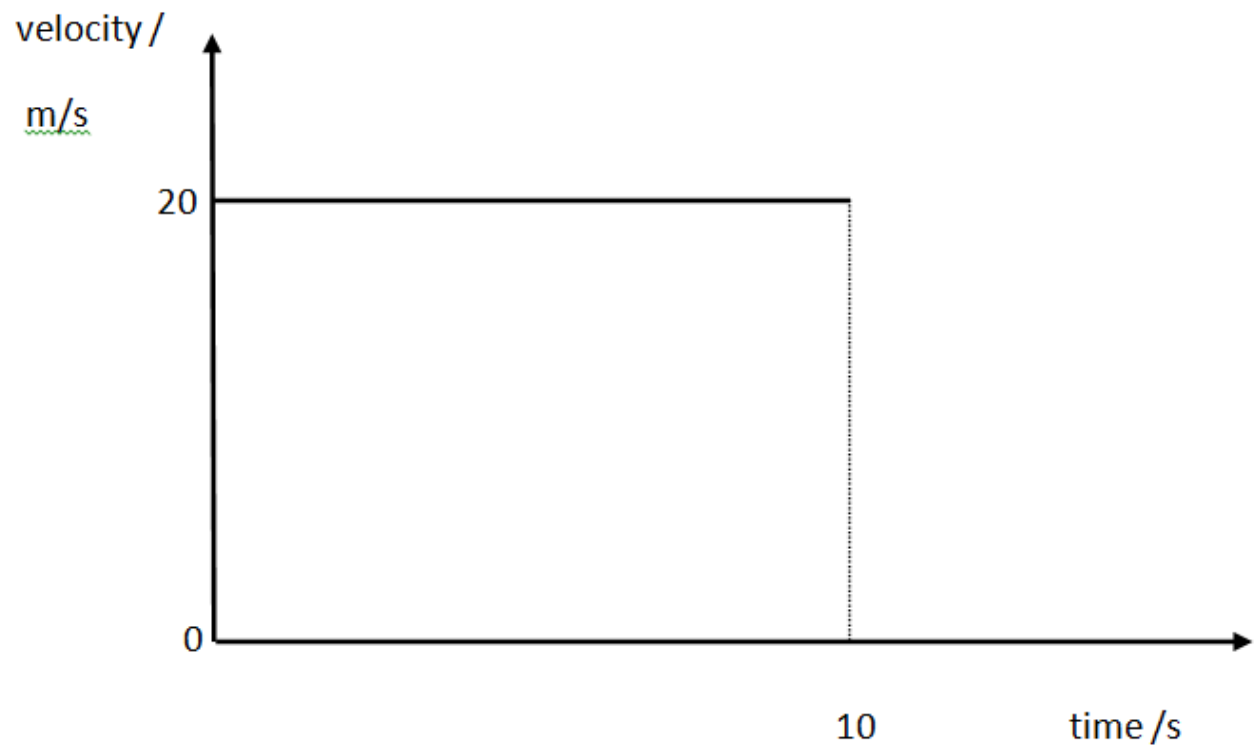
Object B has **constant..... acceleration.....** because the line is a **straight.....** line which goes up.

Object C has **constant..... acceleration.....** **greater than the acceleration of object B**, because the line is a **straight** line which has **steeper.....** gradient **than the line for object B**.

Object D is accelerating with an increasing **rate.....** of acceleration because the line is curving upwards and hence becoming steeper.

<http://www.youtube.com/watch?v=8fmLjSeMCdM>

A simple journey: A car drives at a constant 20m/s for 10s.



What is the distance travelled by an object moving at 20m/s for a time of 10s?

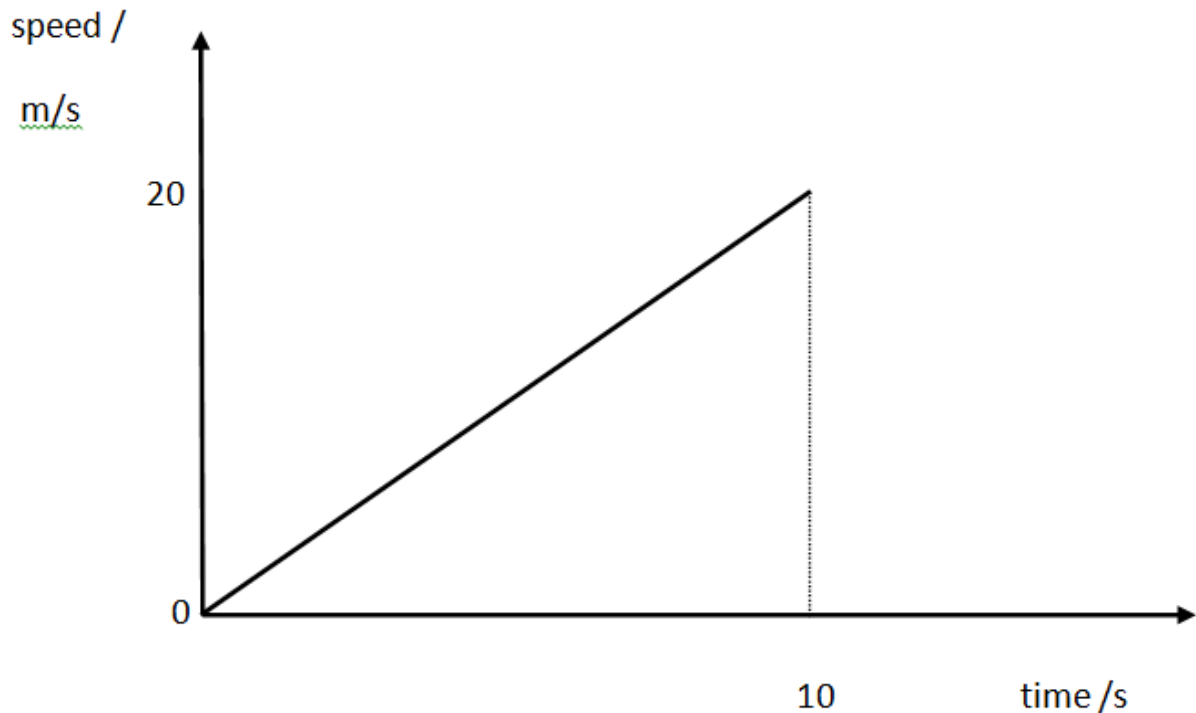
Conclusion

On a Velocity / time graph the gradient..... is the acceleration

On a Velocity / time graph the distance = speed x time under the graph is the distance..... travelled.

Not so simple: What is the distance travelled by an object accelerating uniformly from rest to 20m/s over a time of 10s?

Average speed = $(0 + 20) / 2 = 10 \text{ m/s}$; distance = $10 \text{ m/s} \times 10 \text{ s} = 100 \text{ m}$ (area under graph = $1/2 \times 20 \times 10$).



Extension: (Even harder!)

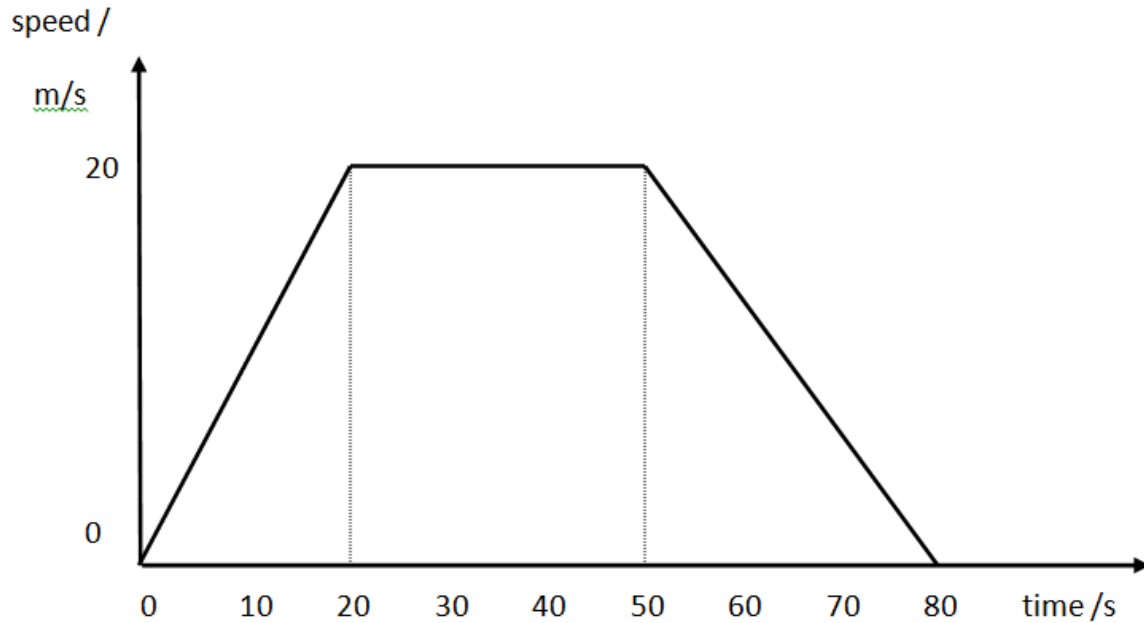
What is the Velocity – Time graph for a bouncing ball?



· Velocity-time graph for a bouncing ball: a series of straight lines all with the same negative gradient (-9.8 m/s^2); at each bounce the velocity changes instantly from negative to positive and the peak speed is smaller each bounce.

Practice Questions

Q1. A car completes the journey below:



1a. What is the distance travelled by the object below over the whole 80s?

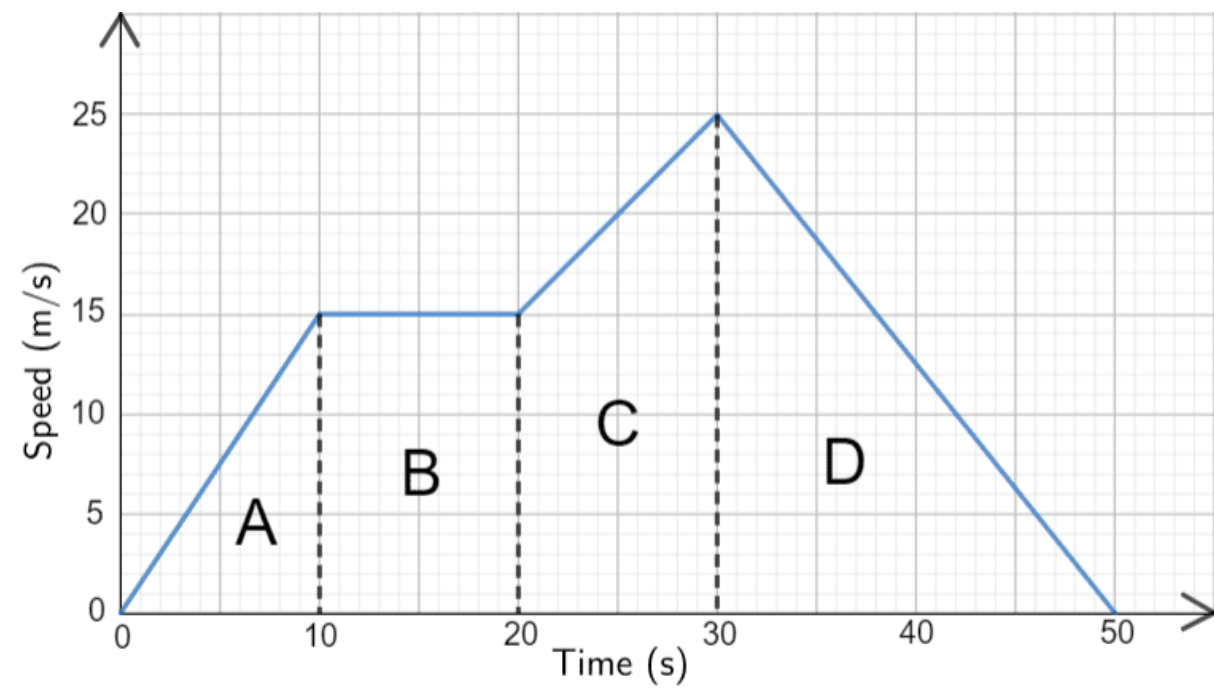
1b. What is the acceleration in the first 20s? $a = \text{gradient} = \text{change in velocity} / \text{time} = (20 - 0) / 20 = 1.0 \text{ m/s}^2$

1c. What is the acceleration between 20s and 50s?

Distance = area under the graph = $(\frac{1}{2} \times 20 \times 20) + (20 \times 30) + (\frac{1}{2} \times 20 \times 30) = 200 + 600 + 300 = 1100 \text{ m}$.

0 m/s² - the velocity is constant between 20 s and 50 s, so the gradient (and therefore the acceleration) is zero.

Q2.



1a. Calculate the total distance travelled in the journey above (3).

1b. Calculate the deceleration in section D.

1c. In which section of the journey is the acceleration the greatest? (1)

- ☒ A. section A
- ☐ B. section B
- ☐ C. section C
- ☐ D. section D

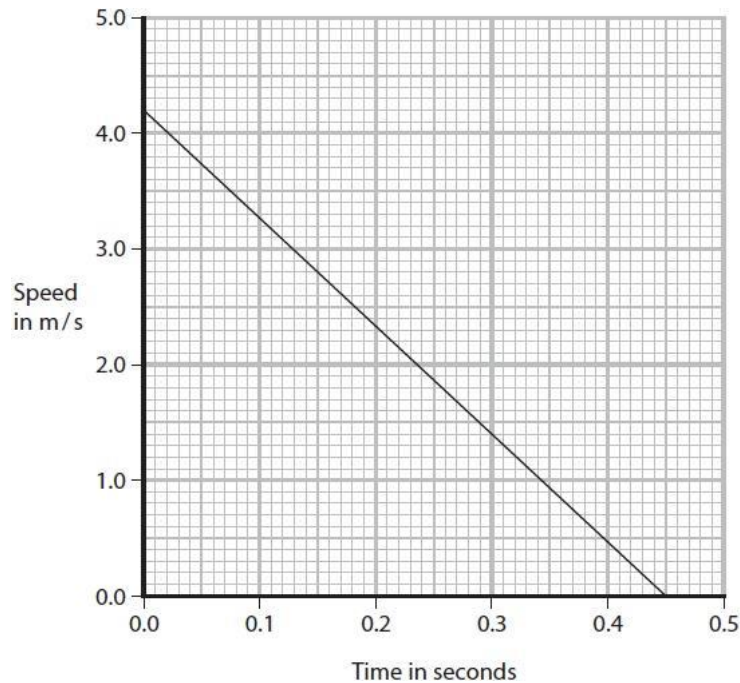
Total distance = total area under the velocity-time graph: add the areas of sections A, B, C and D ($\frac{1}{2} \times \text{base} \times \text{height}$ for triangles, $\text{base} \times \text{height}$ for rectangles) - read the values from the graph.

Deceleration in D = gradient of section D = (final velocity - velocity at start of D) / time for D; the answer is negative, e.g. $(0 - v) / t$.

Q3.

(a) A person throws an object vertically upwards.

The speed-time graph shows how the speed of the object varies from the time it is thrown until reaching its maximum height.



(i) Calculate the acceleration of the object.

(3)

$$\text{acceleration} = -10 \text{ (gradient = } (0 - 4.2) / 0.42 \text{)} \dots \text{ m / s}^2$$

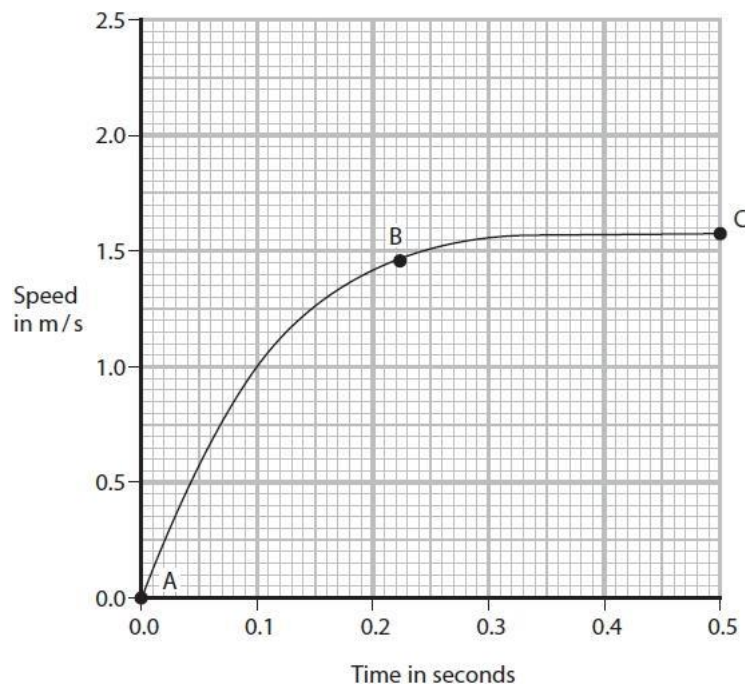
(ii) Calculate the distance the object travels to reach its maximum height.

(3)

$$\text{distance} = 20 \text{ (area under graph = } 1/2 \times 4.2 \times 0.42 \text{)} \dots \text{ m}$$

(b) A different object is dropped from rest and begins to fall.

The graph shows how the speed of this object varies with time.



(i) Give the name of the two forces acting on the object as it falls.

(2)

1 ~~weight~~.....

2 ~~air resistance (drag)~~.....

(ii) Draw arrows on the diagram to show the forces acting on the object at B.

(3)

object

(iii) Explain the shape of the graph from A to C.

You should use ideas about forces to help your answer.

(4)

~~From A, weight makes it accelerate; as speed rises air resistance increases, so resultant force falls. At C.....~~

~~weight = air resistance, resultant force is zero, so speed is constant (terminal velocity).....~~

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 15 marks)

· At B the arrow for weight (downwards) is longer than the arrow for air resistance (upwards), so the resultant force is downwards.

Homework:

P4 Q1-7 and P7 Q1-7 (Hodder book)

If you finish early do the questions in the book (ISLE): P 29 Q2, P35 Q1-2.

Super Extension: (Even harder!)

What is the Velocity – Time graph for a bouncing ball?



· Velocity-time graph: straight parallel lines with the same negative gradient (-9.8 m/s^2); at each bounce the velocity jumps instantly from negative (down) to positive (up) and the peak speed decreases each bounce.

Forces : Lesson 5 : Acceleration Experiment

Learning Objectives

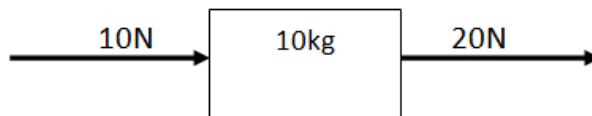
- Forces applied to objects can cause changes of shape (later) or change speed or direction
- I know that acceleration is caused by an **UNBALANCED force**, or **RESULTANT Force**.
- I can recall and use the formula that links unbalanced force, mass and acceleration.
- I am able to use terms such as Research Question, Independent and Dependent variable to describe my investigations
- I can use ICT software such as Excel to present and process experimental data.

1.7	I can plot and explain velocity-time graphs
-----	--

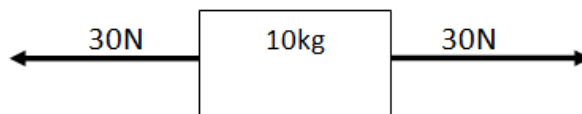
Acceleration

Which of these objects will accelerate? Why?

a.



b.



c.



d. . Consider an object moving in a circle – is it accelerating?

Yes - it is accelerating. Its direction is constantly changing, and acceleration is a change in velocity (velocity is speed in a given direction), so even at constant speed the object is accelerating (centripetal acceleration).

Research Question

What is the relationship between the **RESULTANT force** and the **acceleration** of a trolley?

Independent Variable (Input variable)

Resultant Force

Dependent Variable (outcome variable)

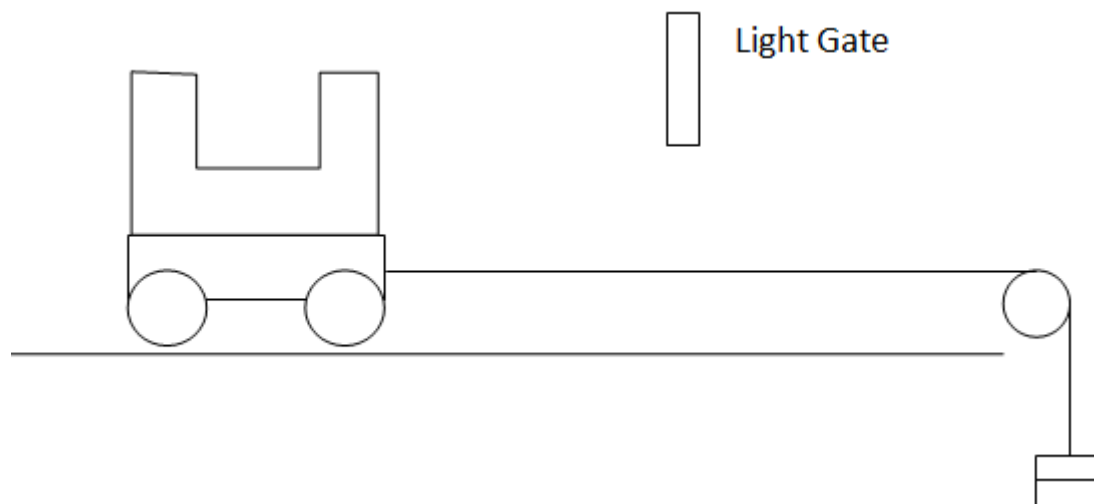
Acceleration of trolley

Control Variables (keep fixed)

- Angle of the slope (horizontal)
- Mass (1kg)
- Start point of trolley
- Medium (air)

Apparatus

Light gates, laptop, spirit level, Ramp, trolley, strong bench pulley and 0.5N weights (50g)



Check out the youtube video of light gates in action on worksheet Forces Lesson 3

Method

1. Use the spirit level to ensure that the slope is horizontal.
2. Accelerate the trolley using a 0.5N force and allow the trolley to accelerate along the ramp and pass through the light gates.
3. Record the acceleration as measured by the computer.
4. **Repeat and calculate an average.**
5. Continue for forces of 1.0N, 1.5N, 2.0N, 2.5N, 3.0N, 3.5N.

Data Collection and Processing

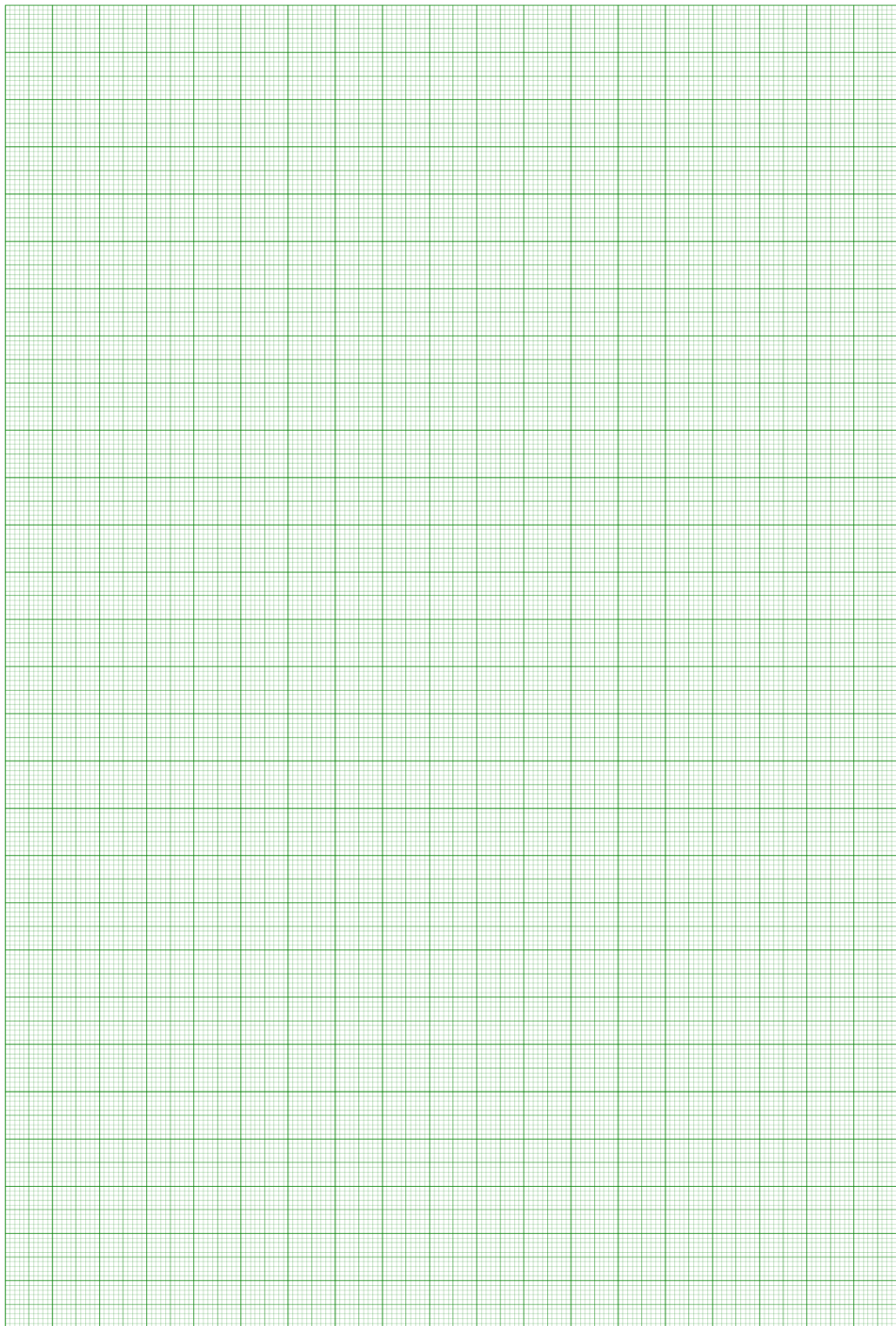
Record results in a suitable manner (using MS Excel will be advantageous)

Resultant Force (N)	Acceleration of the trolley (m/s ²) Repeat 1			Average acceleration (m/s ²)
0.5				
1.0				
1.5				
2.0				
2.5				
3.0				
3.5				

Plot an appropriate graph to answer the research question. Remember the independent variable goes on the x-axis...

Best to go on youtube to check out how to draw a straight line graph on excel/google sheets.

· Plot average acceleration (y-axis) against resultant force (x-axis), join with a best-fit straight line through the origin - showing acceleration is directly proportional to resultant force.



Conclusion

The Graph is a **straight**..... Line that goes through the **origin**.....

This means that Unbalanced Force and acceleration are **directly proportional**.

~~They are directly proportional to one another~~ this means that doubling force would double acceleration.

Extension

Try to calculate the mass of the trolley. Check your value using a top pan balance (can only do if in school)

This lesson we have mainly focussed on experimental techniques. The homework is questions using the **$F = ma$** formula:

Homework:

New Hodder book P17 Q1-8

Old Blue Oxford book; P 39, Q 1-2 and P 50, Q 5

Results for Black Rain Day work:

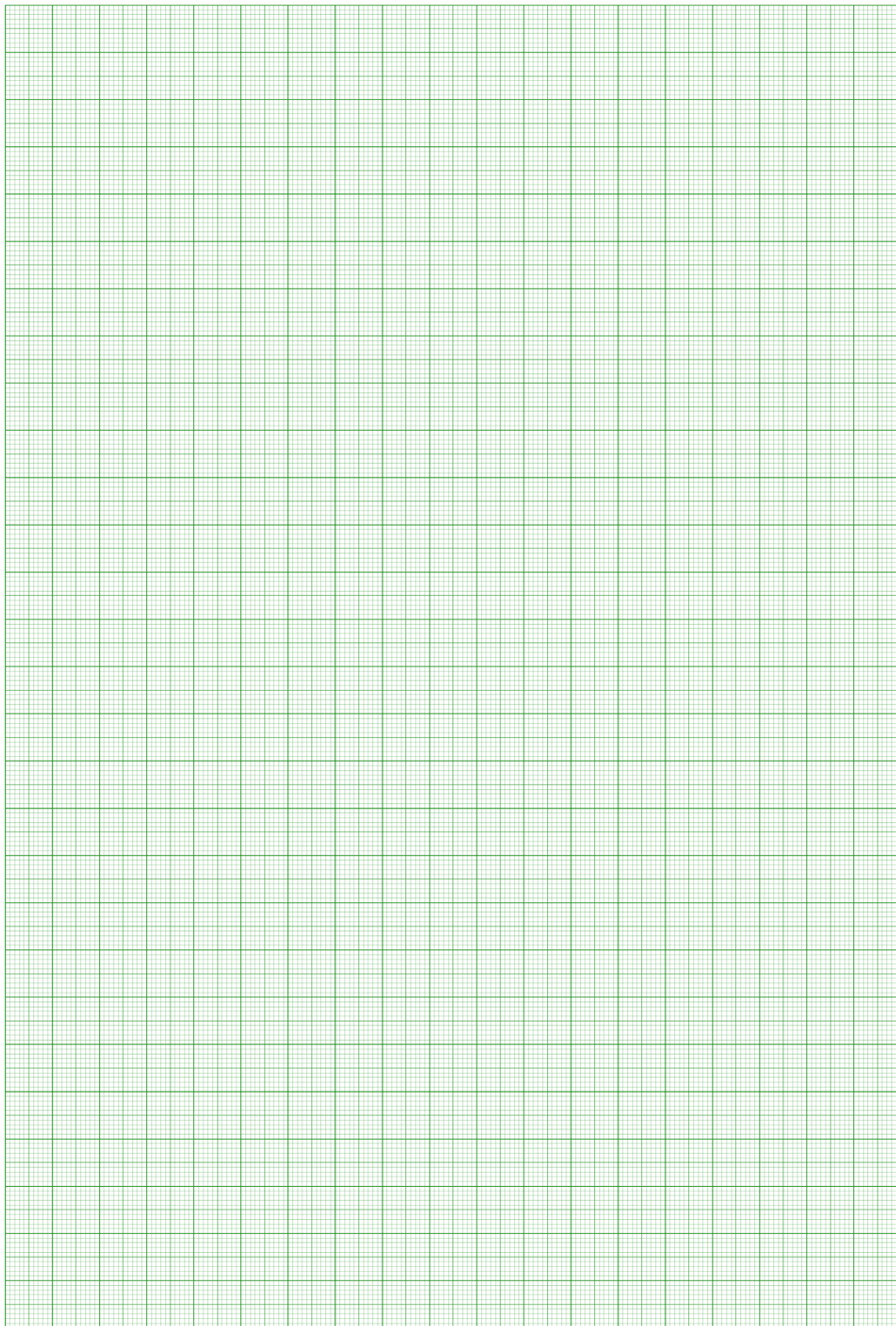
Please use these results to plot an appropriate graph. Identify the Independent Variable (which should be on the x-axis) and the Dependent Variable (y axis). Plot the graph on the provided graph paper, or a piece of graph paper, or you may plot electronically if you do not have either and can use Google sheets.

You will need to work out an average and plot the average (mean) acceleration against Force.

$m = F/a$ (or $m = 1/\text{gradient of the } a\text{-}F \text{ graph}$). Using the results: $m = 1.0 \text{ N} / 0.43 \text{ m/s}^2 = 2.3 \text{ kg}$ (so the trolley plus weights is about 2.3 kg) - check with a top pan balance.

Forces 1.5 (Double Award)

Unbalanced force on trolley +/- 0.1	Acceleration of trolley, repeat 1 +/- 0.01	Acceleration of trolley, repeat 2 +/- 0.01	Acceleration of trolley, repeat 3 +/- 0.01	Mean acceleration of trolley +/-
/ N	/	/	/	/
0.43 1.0	0.43	0.43	0.43	
0.86 2.0	0.87	0.85	0.87	
1.31 3.0	1.30	1.29	1.35	
1.73 4.0	1.77	1.69	1.74	
2.19 5.0	2.17	2.16	2.24	
2.69 6.0	2.69	2.70	2.67	
3.06 7.0	3.03	3.07	3.07	



Forces : Lesson 6 : Acceleration Calculations

Learning Objectives

- I know and can recall three formula involving acceleration
- I can use the formula to calculate: unbalanced force, mass, acceleration, final velocity, initial velocity, time, distance(s).
- I can also calculate the distance moved by an accelerating object.

1.10	I can use the relationship: Use $F = m \times a$, $v = u + at$ and $v^2 = u^2 + 2as$ to link force, mass, acceleration, time and distance. (final speed) ² = (initial speed) ² + (2 x acceleration x distance) or $v^2 = u^2 + 2.a.s$
1.17	I know and can use the formula $v = u + at$ and $v^2 = u^2 + 2as$ - the equations linking acceleration, velocity, time and distance. Force = mass x acceleration or $F = m \times a$ (Newton's Second Law)

Hodder p10

Acceleration = change in velocity/time

Force = mass x acceleration.

$$a = (v-u)/t$$

$$F=ma$$

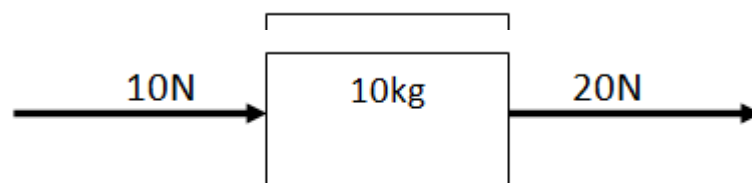
$$a = F / m$$

but what's the 3rd one?? The 3rd relationship is acceleration and distance travelled: $v^2 = u^2 + 2as$ (see p41).

- Acceleration and Force $F = m \times a$: for a given mass, the bigger the resultant force, the greater the acceleration.

Calculate the acceleration of these objects.

a.



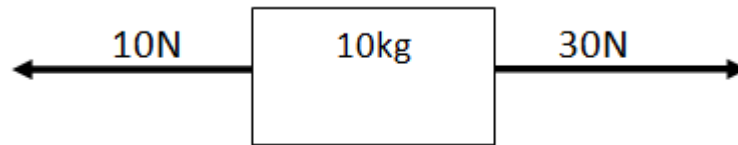
$$F=ma$$

$$30=10a$$

$$a = a = F/m = 30/10 = 3 \text{ m/s}^2 \text{ (units: m/s}^2\text{)}$$

$$(a=F/m)$$

b.



c.



2. Acceleration and Change of Speed

Calculate the following.

a. A car accelerates from 10m/s to 30m/s in 4.7s. Calculate the acceleration.

$$(u = 10\text{m/s}) \quad a = (v-u)/t = (30-10)/4.7 = 20/4.7 = 4.3 \text{ m/s}^2 \text{ (2 sf)}$$

- Read F and m off the diagram, then $a = F/m$. Substitute the values shown, giving a in m/s^2 (show formula, working, answer to 2 sf, units).
- Read F and m off the diagram, then $a = F/m$. Substitute the values shown, giving a in m/s^2 (show formula, working, answer to 2 sf, units).

b. A car accelerates from rest at an acceleration of 2.3m/s^2 for a time of 7.1s . Calculate the final speed. (note rest $u=0\text{m/s}$) $v = u + at = 0 + (2.3 \times 7.1) = 16.33 = 16 \text{ m/s}$ (2 sf)

c. A fighter plane taxis down the runway. After 9.8s of accelerating down the runway with an acceleration of 4.8m/s^2 the fighter plane reaches its takeoff speed of 55m/s . Calculate the initial speed it was taxiing at. $u = v - at = 55 - (4.8 \times 9.8) = 55 - 47.04 = 7.96 = 8.0 \text{ m/s}$ (2 sf)

d. Ussain Bolt runs the 100m in the Olympic Final. He leaves the blocks and accelerates to his maximum speed of 12m/s at an acceleration of 2.1m/s^2 . Calculate the time it took him to reach his maximum speed. $t = (v-u)/a = (12-0)/2.1 = 5.7 \text{ s}$ (2 sf)

3. Acceleration (a) and Distance Travelled (s)

Now we have a new formula:

$$v^2 = u^2 + 2.a.s$$

v = final velocity

u = initial velocity

a = acceleration

s = Displacement (you can just write distance if you want)

a. A car accelerates from 12m/s to 23m/s over a distance of 95m. Calculate the acceleration. $a = (v^2 - u^2) / 2s = (23^2 - 12^2) / (2 \times 95) = (529 - 144) / 190 = 385 / 190 = 2.0 \text{ m/s}^2$ (2 sf)

b. A stone is dropped down a well, which is 20m deep. Calculate the final speed as it hits the water. (Note if dropped $u=0$, and something falling falls at $a=10\text{m/s}^2$)

$v^2 = u^2 + 2as$ - final speed squared = initial speed squared + 2 × acceleration × distance travelled.

$v^2 = u^2 + 2as = 0 + (2 \times 10 \times 20) = 400$, so $v = \text{sq.root } 400 = 20 \text{ m/s}$

C(i). A passenger plane accelerates from rest down the runway. It accelerates at 2.7m/s^2 and reaches its take-off speed of 57m/s . Calculate the length of the runway it requires. $s = (v^2 - u^2)/2a = 57^2/(2 \times 2.7) = 3249/5.4 = 601.7 = 602\text{ m}$ (about 600 m to 2 sf)

C(ii). In the event of an engine failure just before take-off, the same passenger plane can decelerate to a stop at deceleration of 2.0 m/s^2 . Calculate the total length of the runway to safely accommodate the acceleration and the emergency deceleration.

Take-off: $s = 57^2/(2 \times 2.7) = 601.7 \text{ m}$. Stopping: $s = 57^2/(2 \times 2.0) = 3249/4.0 = 812.3 \text{ m}$. Total = $601.7 + 812.3 = 1414 = 1400 \text{ m}$ (1.4 km)

Homework: Complete the following calculations.

Remember the four steps to success:

- Formula
- Working
- Answer (2sf normally)
- Units

$$v^2 = u^2 + 2as$$

Homework:

1. A car decelerates from a speed of 10 m/s to a speed of 4 m/s over a distance of 20 m , owing to a pet dog running across the road. Calculate the deceleration of the car.

$$a = (v^2 - u^2)/2s = (4^2 - 10^2)/(2 \times 20) = (16 - 100)/40 = -84/40 = -2.1 \text{ m/s}^2, \text{ so deceleration} = 2.1 \text{ m/s}^2$$

2. A Formula 1 car accelerates at a constant acceleration of 6.3m/s^2 from an initial speed of 5.5m/s over a distance of 100m . What speed will it reach? $v^2 = u^2 + 2as = 5.5^2 + (2 \times 6.3 \times 100) = 30.25 + 1260 = 1290.25$, $v = \sqrt{1290.25} = 36\text{ m/s}$ (2 sf)

3. A shinkansen accelerates at 0.70m/s^2 for a distance of 1200m and reaches a speed of 52m/s . What is its initial velocity?

4. A long-jumper wants to increase her speed from 2.0m/s to 9.5m/s with an acceleration of 1.7m/s^2 . What length run up should she mark out?

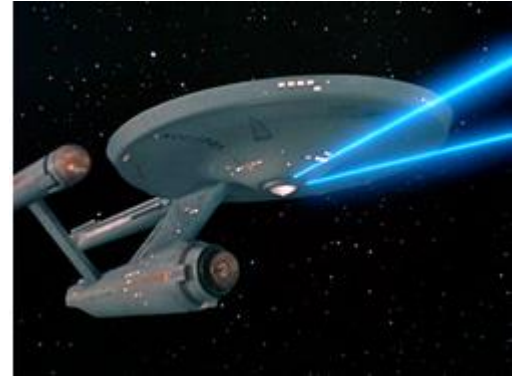
5. A rabbit is hopping towards its burrow at 1.2m/s . The rabbit sees a fox approaching, so the rabbit accelerates at a constant rate towards its burrow. It reaches the burrow at a speed of 15m/s in a time of 4.4s . Calculate the distance travelled to the burrow.

$$u^2 = v^2 - 2as = 52^2 - (2 \times 0.70 \times 1200) = 2704 - 1680 = 1024, u = \sqrt{1024} = 32\text{ m/s}$$

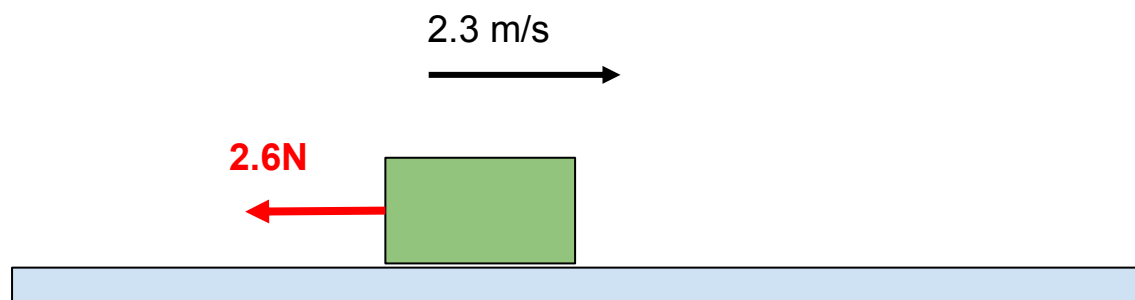
$$s = (v^2 - u^2) / 2a = (9.5^2 - 2.0^2) / (2 \times 1.7) = (90.25 - 4.0) / 3.4 = 86.25 / 3.4 = 25\text{ m}$$
 (2 sf)

$$a = (v - u) / t = (15 - 1.2) / 4.4 = 3.14\text{ m/s}^2; s = (u + v) / 2 \times t = (1.2 + 15) / 2 \times 4.4 = 8.1 \times 4.4 = 35.6 = 36\text{ m}$$
 (2 sf)

6. The starship Enterprise NCC1701 is having a lot of trouble with Klingons. Captain Kirk orders Scotty to accelerate from $290,000,000 \text{ m/s}$ with an acceleration of 7000 m/s^2 . They have 900 billion metres of clear space ahead before they will be within range of the Klingon fleet. Will they escape to safety by entering warp speed (faster than light)?



Extension:



A 2.7 kg block is sliding across a table at an initial speed of 2.3 m/s . The force of Friction on the block is 2.6 N

Qa. The force of friction always acts *with* / against the direction of motion. (1)

Qb. Calculate the deceleration of the block. (2)

Qc. Calculate the distance the block slides before stopping. (3)

Use $v = s \div t$ and $a = (v - u) \div t$, converting units first ($1 \text{ km} = 1000 \text{ m}$, $1 \text{ h} = 3600 \text{ s}$, $1 \text{ m/s} = 3.6 \text{ km/h}$).

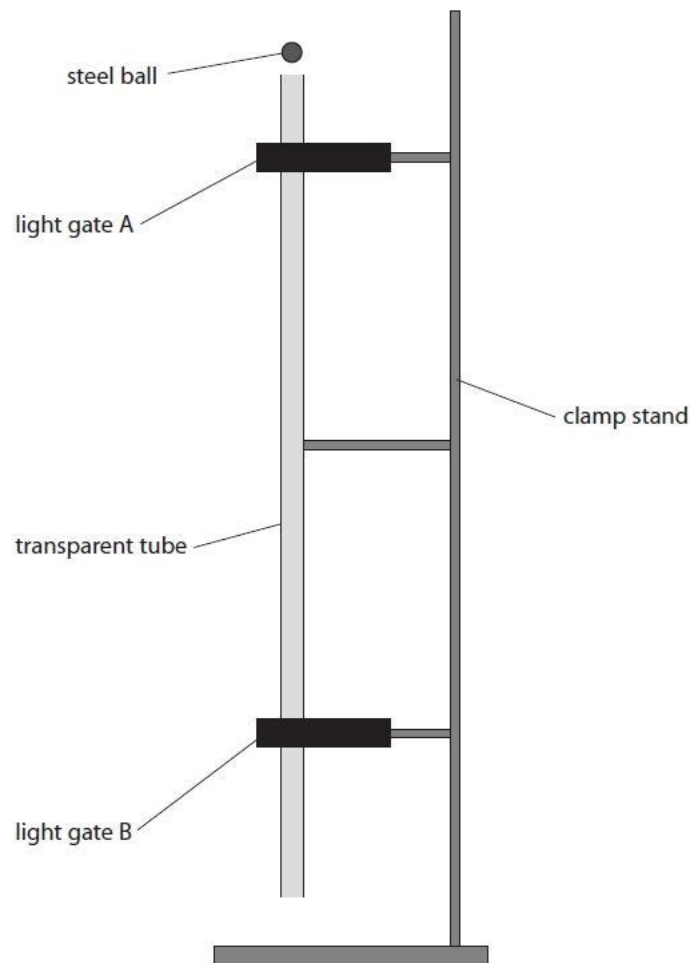
$v^2 = u^2 + 2as = (2.9 \times 10^8)^2 + (2 \times 7000 \times 9 \times 10^{11}) = 8.41 \times 10^{16} + 1.26 \times 10^{16} = 9.67 \times 10^{16}$, $v = 3.1 \times 10^8 \text{ m/s}$. This is greater than the speed of light ($3.0 \times 10^8 \text{ m/s}$), so yes - they would reach warp speed (and escape).

$F = ma$, so $a = F/m = 2.6/2.7 = 0.96 \text{ m/s}^2$. Deceleration = 0.96 m/s^2 (the 2.6 N friction opposes motion).

$v^2 = u^2 + 2as$, so $s = (v^2 - u^2)/2a = (0 - 2.3^2)/(2 \times -0.96) = -5.29/-1.92 = 2.7 \text{ m}$ (2 sf)

Practice Questions

A student does an experiment to determine the acceleration due to gravity, g .
The diagram shows the apparatus used.



This is the student's method.

- connect both light gates to a data logger
- drop a steel ball from rest at the top of the transparent tube
- record the speed of the ball at each light gate
- record the time taken for the ball to fall from light gate A to light gate B

(a) The box shows the data recorded by the data logger.

speed at A = 1.45 m/s
speed at B = 4.20 m/s
time from A to B = 0.286 s

- (i) Show that the acceleration of the steel ball is approximately 9.6 m/s^2 .

(2)

$a = (v - u) / t$. Substitute the two light-gate speeds and the time from the data logger,
e.g. $(2.4 - 1.2) / 0.125 = 9.6 \text{ m/s}^2$.

acceleration = 9.6 m / s^2

- (ii) Explain why the student's value for the acceleration of the steel ball is lower than the accepted value for the acceleration due to gravity, g .

(2)

~~Air resistance acts upwards on the ball as it falls, opposing its motion.~~.....

~~This upward drag makes the resultant force smaller than the weight, so $a = F/m$ is less than g .~~.....

.....

.....

- (iii) Calculate the distance between the light gates.

(3)

distance = 0.225 m

- (b) The student changes the distance between the light gates by varying the position of light gate B.

The student measures the time taken for the steel ball to fall from light gate A to light gate B when the light gates are different distances apart.

The table shows the student's results.

Distance between light gates in m	Time taken in s
0.10	0.058
0.20	0.103
0.30	0.141
0.40	0.175
0.50	0.205
0.60	0.233
0.70	0.260

$s = \text{average speed} \times \text{time} = (u + v)/2 \times t = (1.2 + 2.4)/2 \times 0.125 = 0.225 \text{ m}$ (or $s = ut + 1/2 at^2$).

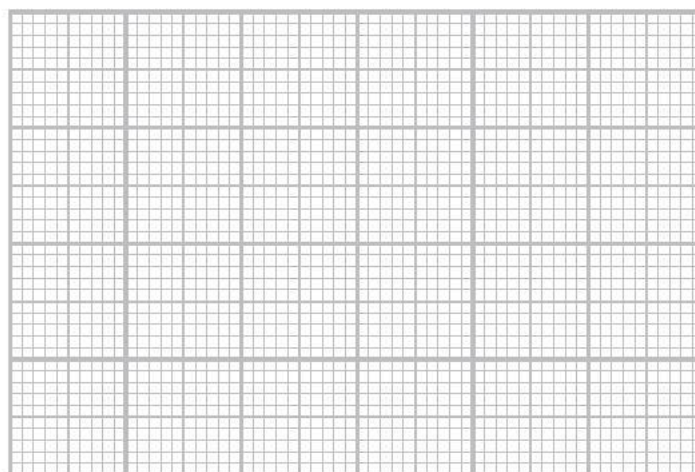
(i) Plot a graph of the student's results on the grid.

(2)

(ii) Draw the curve of best fit.

(1)

Distance between
light gates
in m



Time taken in s

(iii) Explain how the graph shows that the steel ball is accelerating as it falls.

(3)

~~The graph is a curve whose gradient (steepness) gets bigger as time goes on.~~
~~Gradient of a distance-time graph = speed; so an increasing gradient means the speed is increasing; i.e. the ball is accelerating.~~

(Total for question = 13 marks)

Q5.

A driver of a car sees an obstruction in the road ahead and must stop the car.

(a) (i) State the formula linking average speed, distance travelled and time taken.

(1)

(ii) A car travels at 21 m / s.

The driver's reaction time is 0.14 seconds.

Calculate the distance travelled by the car during the driver's reaction time.

(2)

- Plot distance / m on the y-axis against time / s on the x-axis;
all points from the table plotted, then joined with a smooth curve.
- Draw one smooth curve of best fit through the plotted points (not straight lines joining the points).

distance = 2.94..... m

average speed = distance travelled / time taken ($v = s / t$)

distance = speed x time = $21 \times 0.14 = 2.94$ m

(b) The car experiences a braking force of 7600 N.

The car has a mass of 1200 kg.

(i) State the formula linking force, mass and acceleration. $\text{force} = \text{mass} \times \text{acceleration}$ ($F = m a$)
(1)

(ii) Calculate the acceleration of the car.

(2)

acceleration = 6.33..... m / s²

(iii) Calculate the braking distance travelled as the speed of the car is reduced from 21 m/s to 0 m/s.

(3)

distance = 35..... m

(Total for question = 9 marks)

$$a = F / m = 7600 / 1200 = 6.33 \text{ m/s}^2 \text{ (the braking force opposes motion, so the car decelerates).}$$
$$v^2 = u^2 + 2as \rightarrow 0 = 21^2 + 2 \times (-6.33) \times s \rightarrow s = 441 / 12.66 = 34.8 \text{ m}$$

Forces : Lesson 7 : Terminal Velocity

Learning Objectives

- I know what terminal velocity is!
- I know that weight is a force and the unit for weight.
- I can recall and use the formula that links weight and gravity.
- I can describe the key forces and how they change during, say, a parachute jump.
- I can draw a velocity-time graph for, say, a parachute jump.

1.18	I know and can use the formula Weight = mass x gravitational field strength or $W = m \times g$
1.21	I can describe the forces acting on falling objects and explain why they reach terminal velocity

P 18-19

Weight is a ~~force~~..... measured in ~~newtons~~..... (N)

Weight

Mass is an amount of ~~matter~~..... . Measured in

~~kilograms (kg)~~..... (.....)

Eg. An astronaut goes into space, she can be ~~weightless~~....., but

her ~~mass~~..... will stay the same.

$$\text{Weight} = \text{mass} \times \text{gravity}$$

or

$$W = m \times g \quad (F=ma)$$

weight = mass $\times$ gravitational field strength, $W = m \times g$ (W in N, m in kg, g in N/kg; $g = 9.8$ N/kg on Earth).
Weight acts down, air resistance up; drag increases with speed until drag = weight, resultant force = 0, so the object falls at terminal velocity.

Eg. Jeremy has a mass of 74kg. What is his weight? $W = m \times g = 74 \text{ kg} \times 10 \text{ N/kg} = 740 \text{ N}$

Weight = mass x gravity

$$= 74 \dots\dots \times 10 \dots\dots$$

$$= 740 \dots\dots \text{ N}$$

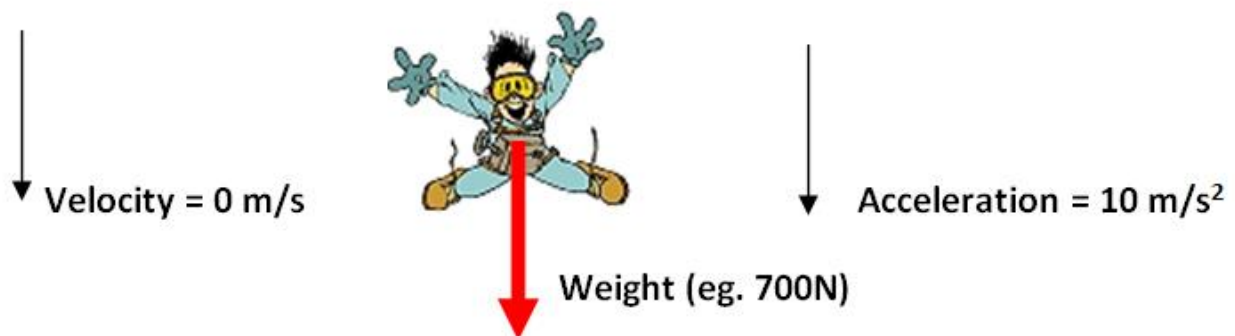
Here on earth Gravity is approx **10N/kg or 10 m/s²**. We call this the **strength of gravity** or the **acceleration due to gravity (called 'g')** or the **gravitational field strength**.

http://www.youtube.com/watch?v=5C5_dOEyAfk

https://youtu.be/E43-CfukEgs?si=B_wDLHkKTe_PCFLB

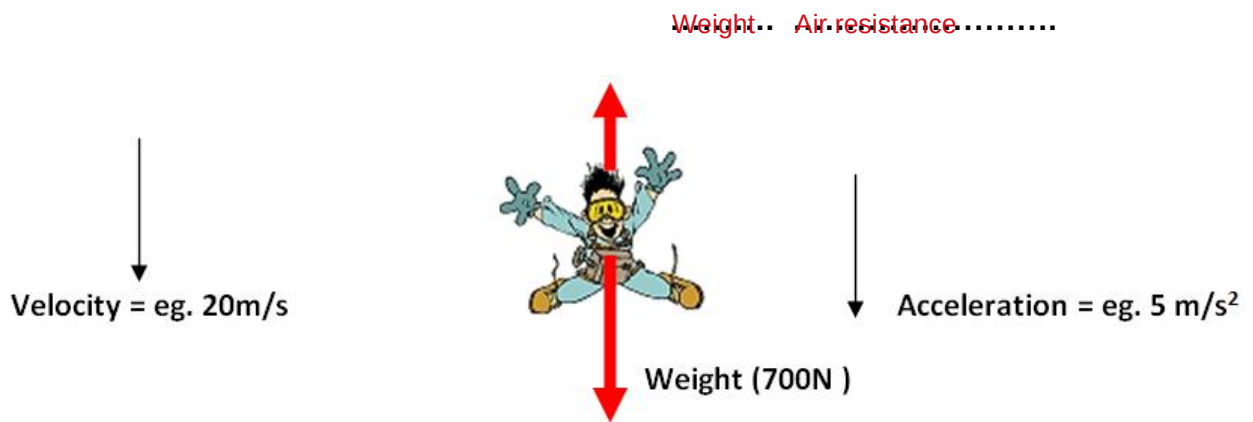
Consider an object, such as a skydiver falling from rest.....

1. The **initial moment** of the jump (from a stationary hot air balloon)



The parachutist is **accelerating**..... because there is an **unbalanced**..... force

2 A few seconds later



There now is some **air..... resistance.....** caused by air particles hitting your body as you fall through the air. Because of air resistance the

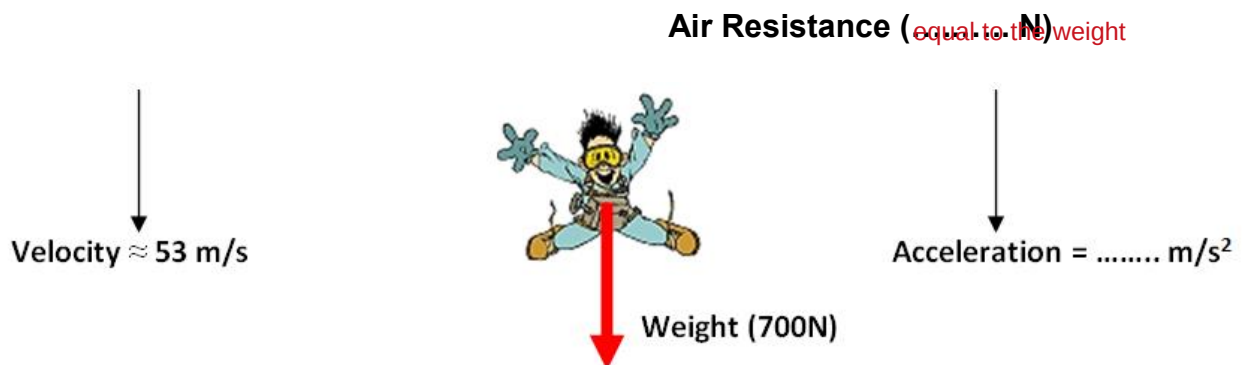
resultant..... force is reduced. The skydiver still has an

acceleration..... (now smaller than 10m/s^2) because there is still an

unbalanced..... FORCE

3. Approx 20 seconds later.....(fill in the gaps and draw the missing arrow)

· Draw the missing arrow: an upward air-resistance arrow the same length as the downward weight arrow, so the two forces balance.

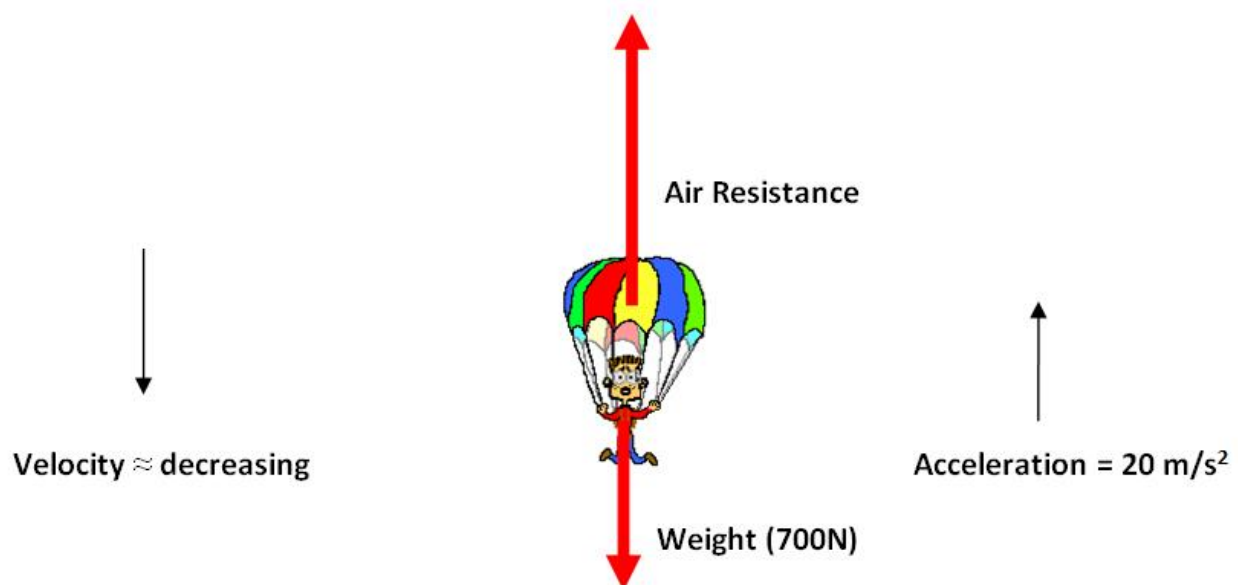


As the velocity has increased, the air resistance has also increased .

THE FORCES ARE NOW balanced..... !!!!!!!

The acceleration is ZERO the skydiver now at terminal..... VELOCITY.

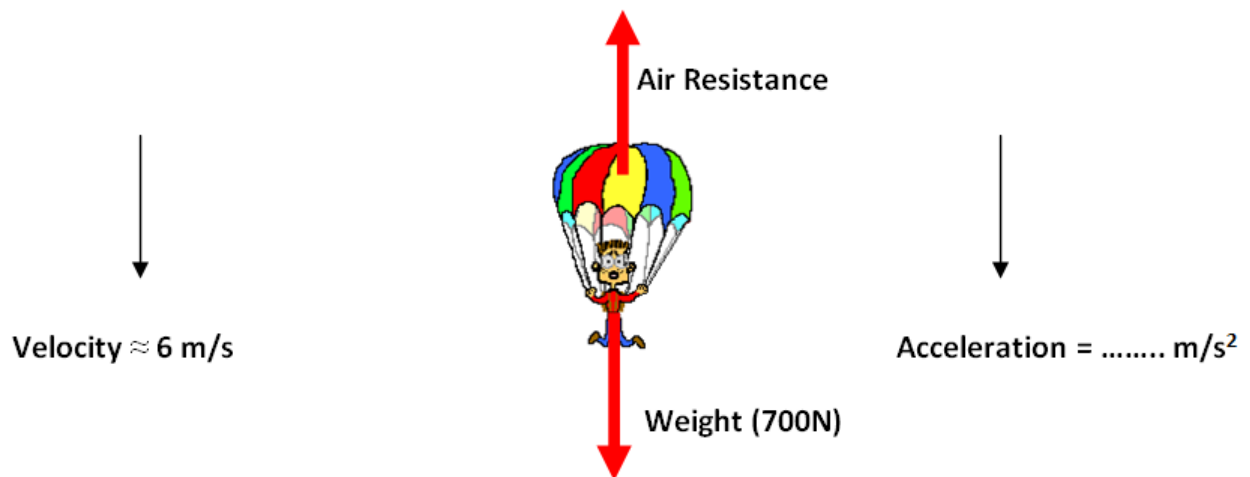
4. Opening the parachute.....



The parachute has a much greater surface..... area..... and this causes an increase in air..... resistance...

The air resistance is greater than the weight so there is an upward..... FORCE causing the parachutist to decelerate..... (there is actually an upwards acceleration)

5. A few seconds after opening the parachute.....



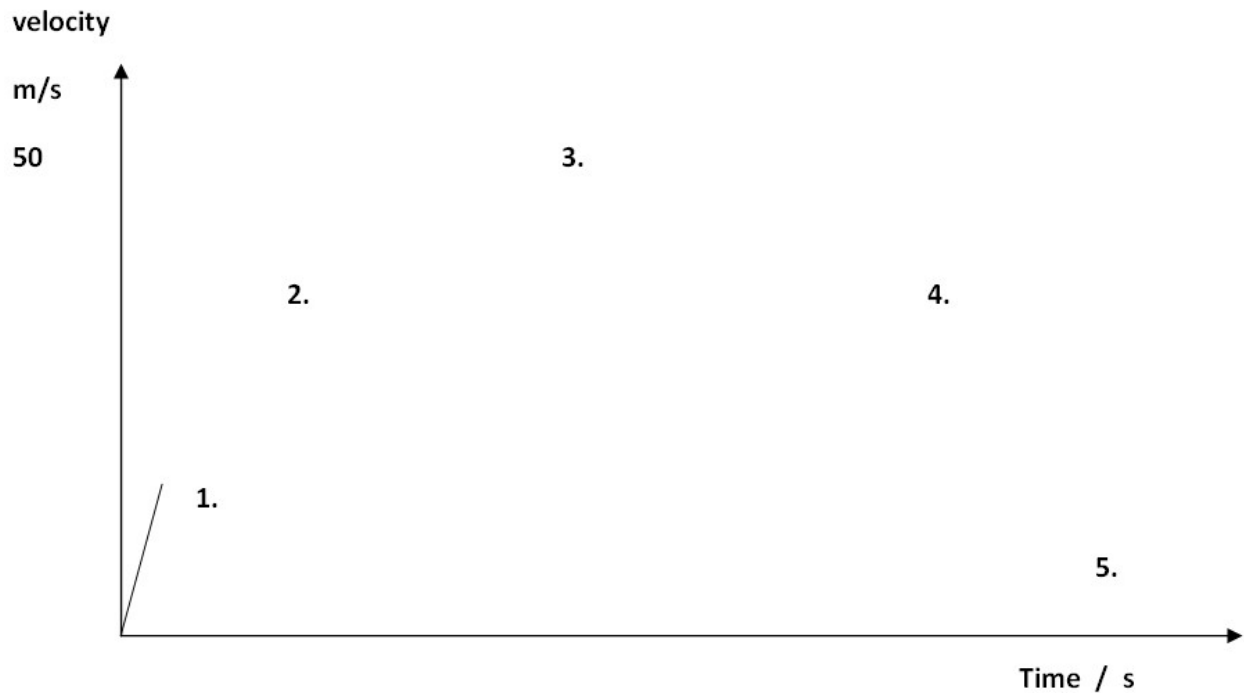
As the parachutist decelerates the air resistance will **decrease**..... until air resistance and

weight are again **balanced**..... !!! The acceleration is **zero**..... the skydiver now

at a new **SLOWER** **terminal**..... velocity

Q - Sketch a Velocity-Time graph for a free-fall parachutist

· Sketch: curve from 0 flattening to a horizontal line at ~ 50 m/s after ~ 20 s; when the parachute opens the line drops steeply to a lower horizontal line (new, slower terminal velocity), then falls to 0 on landing.



[This would not happen on the moon because there is no air on the moon!](#)
[\(You Tube link\)](#)

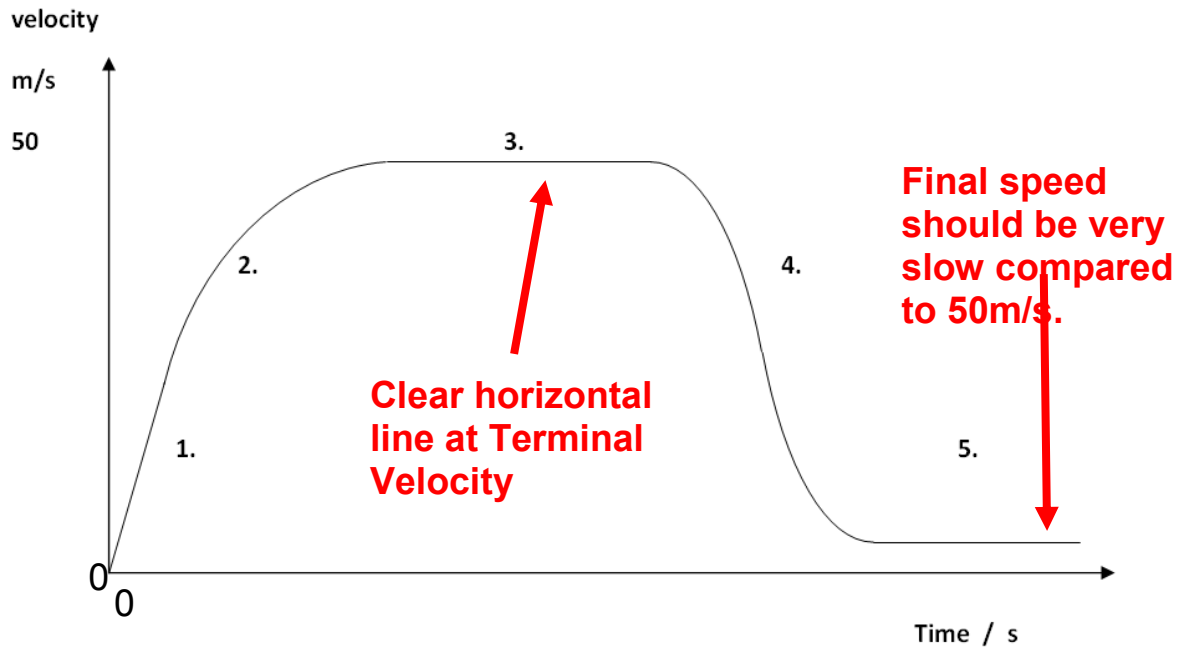
[Peregrine Falcon - extreme streamlining](#)

Homework:

Hodder P21 Q6

Blue book P37, Q 1-3.

Model answer to question above:



Newton's Laws of Motion

N I – Objects remain at rest or at uniform velocity unless acted upon by a resultant force.

N II – The acceleration of an object is directly proportional to the resultant force.

The simple formula for this is ~~force~~ = mass x acceleration

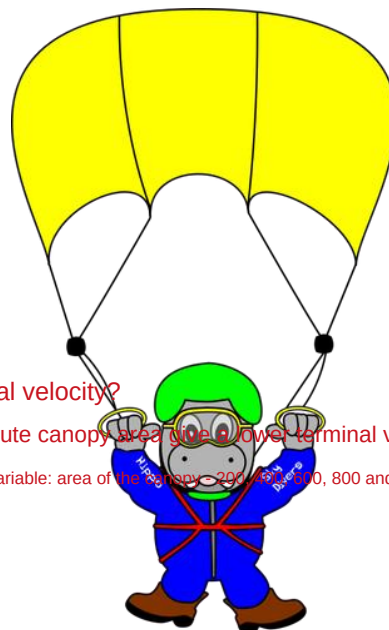
$$\mathbf{F = m a}$$

N III – For every force upon an object there is an equal and opposite force upon another object.

Extension: Design an Experiment to Investigate a factor that affects the Terminal Velocity of a Parachute.

Experimental Design – Checklist:

1. **Title** title: How does the area of a parachute affect its terminal velocity?
2. **Research Question** Research question: Does a larger parachute canopy area give a lower terminal velocity?
3. **Independent Variable (include numerical values)** Independent variable: area of the canopy – 200, 400, 600, 800 and 1000 cm².
4. **Dependent variable (the terminal velocity)**
5. **Control Variables (list and give a numerical value to “fix/control” the variable)**
6. **Prediction (explain using science)**
7. **Apparatus (list plus a labeled diagram)**
8. **Method (numbered step by step procedure)**
 1. Give numerical values for the independent variable
 2. Repeat readings
 3. Calculate an average
 4. Plot an appropriate graph



Dependent variable: terminal velocity, found by timing the last 1 m of the fall ($v = \text{distance} / \text{time}$).

Control variables: same load mass (20 g) and shape, same drop height (3 m), same canopy material, same room with no draughts.

Prediction: the larger the canopy area, the greater the air resistance at any speed, so terminal velocity is reached at a lower speed (drag = weight).

Apparatus: plastic sheet canopies, string (4 x 30 cm), 20 g mass, metre rule, stopwatch, clamp stand; labelled diagram of canopy, strings and load.

Method: cut each canopy, tie strings to the 20 g mass, drop from 3 m, time each fall 3 times, average, then plot terminal velocity against canopy area.

Forces: Lesson 8 :Stopping Distance

Learning Objectives

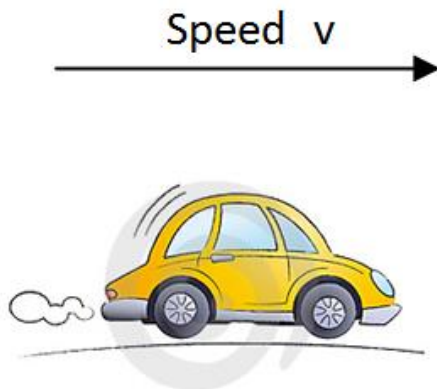
- I know both phases that occur when a moving car comes to a stop
- The brakes on a car work by Friction, which is a force that opposes motion
- I know what conditions make it more difficult to stop and hence more dangerous for driving
- I know the formula for **Kinetic Energy** and **Work Done** and can apply them to a braking car

1.19	I know that stopping distance = thinking distance + braking distance
1.20	I can describe the factors affecting stopping distance including speed , mass , road condition and reaction time

Hodder P20-21

Road safety is a very important issue. Accidents will occur if cars cannot stop in a short enough distance.

Consider a car driving down the road at constant speed v :



Suddenly a small boy runs out into the road. What is the first thing that happens?

The driver must **react**..... . This takes **time**..... !

We can call this time the : **reaction**..... **time** or **thinking**..... **time**!

The car is still moving at this time. The distance the car moves in this time is

called the **thinking**..... **Distance** or **reaction**..... **distance**.

stopping distance = thinking distance + braking distance.

Factors: speed, vehicle mass, road/weather conditions, tyre and brake condition, and the driver's reaction time (alcohol, drugs, tiredness, distraction).

Practical – What is my reaction time?

Catch a falling ruler,

The ruler should be released with no prior warning. Note the distance that the ruler fell.

We can convert the distance the ruler falls into time using this formula:

(Mine 13cm = 0.13m) $a = 10 \text{ m/s}^2$ $t = \text{reaction time}$

$$\text{Distance} = \frac{1}{2} a t^2$$

Example of thinking Distance:

<https://www.youtube.com/watch?v=oal-vBFmnRk>

Eg1 . My reaction time is 0.6 seconds. I am driving at the speed limit in Hong Kong of 50km/hr. 50km/hr is 14m/s. Use 14m/s in your calculation.

What is my thinking distance? v, t, s speed, distance, time

Of course I was concentrating very intensely, on a typical journey the reaction time may be 1.5s:

<http://www.visual-expert.com/Resources/reactiontime.html>

Ruler-drop test: a partner drops a ruler between your fingers; reaction time $t = \sqrt{(2s \div g)}$. E.g. caught at 20 cm $\cdot t = 0.2 \text{ s}$. Repeat and average.

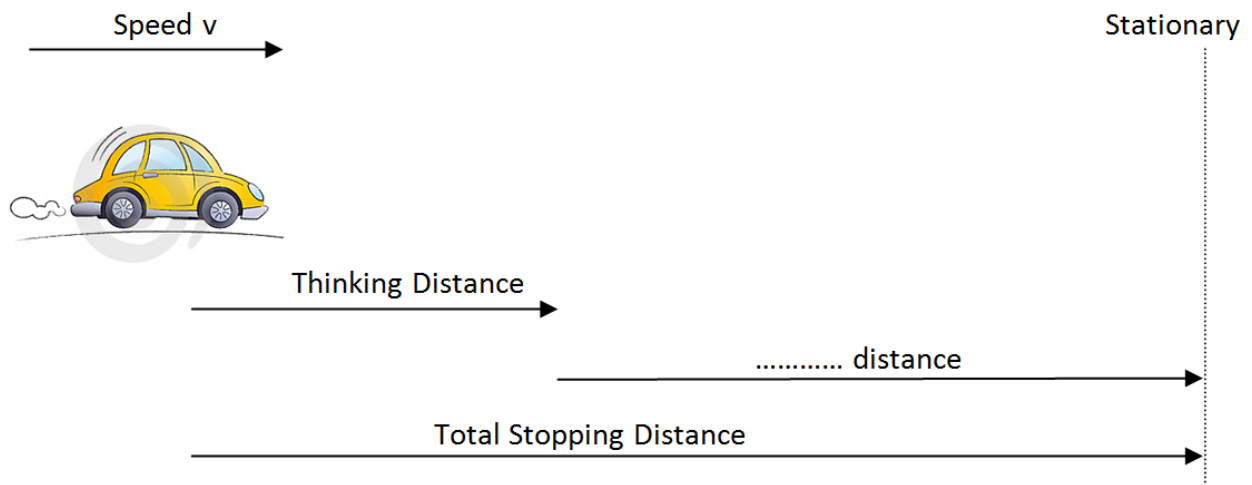
$$\begin{aligned} t &= \text{square root of } (2 \times \text{distance} / a) = \text{square root of } (2 \times 0.13 / 10) \\ &= \text{square root of } 0.026 = 0.16 \text{ s} \end{aligned}$$

thinking distance = speed $\times$ time = $14 \times 0.6 = 8.4 \text{ m}$

Eg 2. My reaction time is 1.5 seconds. I am driving at the speed limit in Hong Kong on the Tolo Harbour Highway of 100 km/hr. 100 km/hr is 28m/s. Use 28m/s in your calculation.

What is my thinking distance? Thinking distance = speed x reaction time = 28 m/s x 1.5 s = 42 m

What is the Next stage? Braking stage - the brakes are applied and friction slows the car until it stops; this distance is the braking distance.



After the thinking time the brakes are now applied and the car starts to brake.

Brakes create friction which slows down the car. Friction always opposes motion.

The distance it now takes for the car to brake to a stop is the **braking**.....
DISTANCE.

$$\text{Total Stopping Dist} = \text{Thinking Dist} + \text{Braking Dist}$$

Braking Distance and Energy.

When moving, the car has **kinetic**..... **energy**.

The KE is transferred..... by the brakes into thermal..... energy .

The KE transferred is the work..... done by the brakes.

$$KE = \frac{1}{2} m v^2$$

KE = Kinetic Energy (J)

m = mass (kg)

Velocity= speed (m/s)

And

$$\text{work done} = \text{Force} \times \text{distance}$$

(J)

(N)

(m)

$$KE = \text{Work done by brakes}$$

$$\frac{1}{2} m v^2 = \text{Force} \times \text{distance}$$

The force is the force from the brakes (the force of friction) and the distance is the braking distance.

Example Questions

Q1a. What is the braking distance for a family car moving at 17m/s?

Mass = 900kg

Force $\approx$ 8000N

$$\frac{1}{2} m v^2 = \text{Force} \times \text{distance}$$

Q1b If the reaction time of the driver was 1.0s, calculate the stopping distance.

What happens if the speed is doubled?

$$\frac{1}{2} m v^2 = \text{Force} \times \text{distance} \rightarrow 0.5 \times 900 \times 17^2 = 130\,050 \text{ J}; d = 130\,050 / 8000 = 16.3 \text{ m}$$

$$\text{Thinking distance} = 17 \times 1.0 = 17 \text{ m. Stopping distance} = 17 + 16.3 = 33.3 \text{ m}$$

Doubling the speed quadruples the braking distance ($KE = \frac{1}{2} m v^2$ so KE is proportional to v^2) and doubles the thinking distance, so stopping distance is much greater.

Q2. A car is driving down the road at a constant speed of 23m/s.

- a. The driver's reaction time is 0.80s. Calculate the thinking distance.

Thinking distance = speed x reaction time = $23 \times 0.80 = 18.4 \text{ m}$ (18 m)

- b. The mass of the car is 950kg and the brakes exert a force of 8700N.

Calculate the braking distance. $\frac{1}{2} m v^2 = F \times d \rightarrow 0.5 \times 950 \times 23^2 = 251\,275 \text{ J}; d = 251\,275 / 8700 = 28.9 \text{ m}$

- c. What is the total stopping distance? Total stopping distance = $18.4 + 28.9 = 47.3 \text{ m}$

Stopping Problems:

What problems can you think of for a car that needs to make an emergency stop?

Eg. What might make conditions worse?

If a car's stopping distance is longer than expected, the car may not react in time, and the road/tyres may give less

<u>Problem</u>	<u>Effect on Thinking Distance</u>	<u>Effect on Braking Distance</u>
Wet/icy road: thinking unchanged, braking distance longer (less friction). Tired/drunk: longer reaction time, so longer thinking distance, braking unchanged. Worn tyres/brakes: less friction, so longer braking distance.		

<http://www.sdt.com.au/safedrive-directory-STOPPINGDISTANCE.htm>

How can a driver be distracted?

Homework:

Hodder P21 Q7

Other homework Blue Book – P41, Q1 & 3 & 5, and P79, Q 1-2, P83, Q 2-3

Using a mobile phone, texting, talking to passengers, changing the radio or sat-nav, eating or drinking, tiredness, or looking at something outside the car.

Summary of Stopping Related Problems

<u>Problem</u>	<u>Effect on Thinking Distance</u>	<u>Effect on Braking Distance</u>
Driver is Tired, Drunk, or on Drugs	Slower reactions, longer reaction time and longer thinking distance	None (the brakes are not affected by the mental state of the driver)
Distracted Driver (eg. mobile phone)	Slower reactions, longer reaction time and longer thinking distance	None
Road is wet or icy	None	There is less friction and braking distance will be longer
Tyres are worn or brake pads are worn or badly adjusted.	None	There is less friction and braking distance will be longer
Car is overloaded	None	mass is too high, therefore KE is too great and braking distance is longer
Speeding	Yes - increased	KE is greatly increased and braking distance is greatly increased. Doubling speed..... will quadruple the braking distance.

Forces : Lesson 9 & 10 : Hooke's Law

Learning Objectives:

- How do I investigate a research question?
- What is Hooke's Law?
- What is Hookian behaviour?
- What is the meaning of the spring constant?
- I can design an experiment to investigate the relationship between the force on a spring, wire or rubber band and its extension

Plan a method with independent, dependent and control variables; repeat and average readings; plot a graph and evaluate the uncertainties.

1.22	Practical: I can investigate how extension varies with applied force for helical springs, metal wires and rubber bands
1.23	I know that the initial linear region of a force-extension graph is associated with Hooke's Law
1.24	I can describe elastic behaviour as the ability of a material to recover its original shape once deforming forces are removed

Hodder P22-25

Research Question for metal spring:

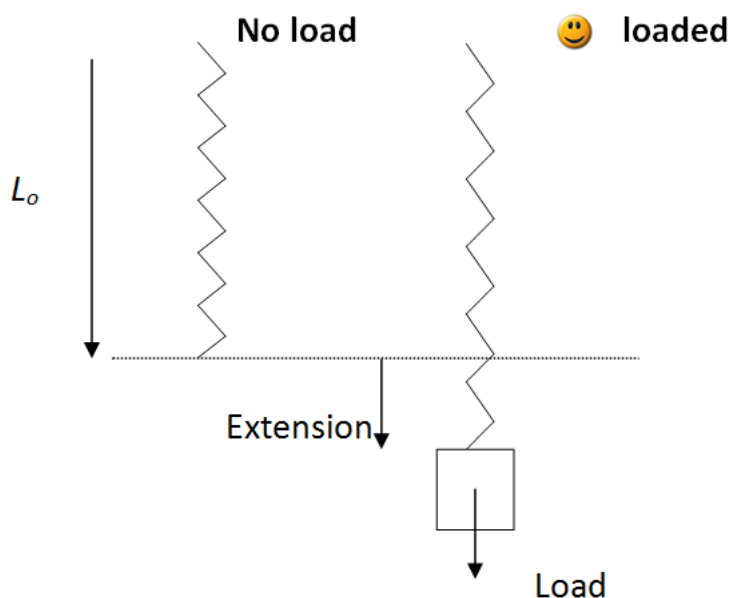
What is the relationship between the **Force** and the **extension** for a metal helical spring?

§ Independent variable is the **load (force) / Newtons**

§ Dependent variable is the **extension / cm**

§ **What are the control variables?**

Apparatus



Think about making your experimental technique better.....

- **How can I make my data more accurate?**
- **How do I make my data more reliable?**

Results

<u>Force / N</u>	<u>Extension</u> <u>/ cm</u> <u>Repeat 1</u>	<u>Extension</u> <u>/ cm</u> <u>Repeat 2</u>	<u>Average</u> <u>Extension</u> <u>/ cm</u>
0	0.0	0.0	0.0
1			
2			
3			
4			
5			
6			
7			

Read the ruler at eye level next to a pointer at the bottom of the spring to avoid parallax, and measure to the nearest mm.

Repeat each measurement three times and calculate a mean; take readings at the same force intervals and check for anomalous results.

· Example model data (Force/N -> average extension/cm): 0->0.0, 1->1.6, 2->3.2, 3->4.8, 4->6.4, 5->8.0, 6->9.6, 7->11.4. Straight line through the origin up to about 5 N.

(use an excel spreadsheet or a Googledoc to share the data)

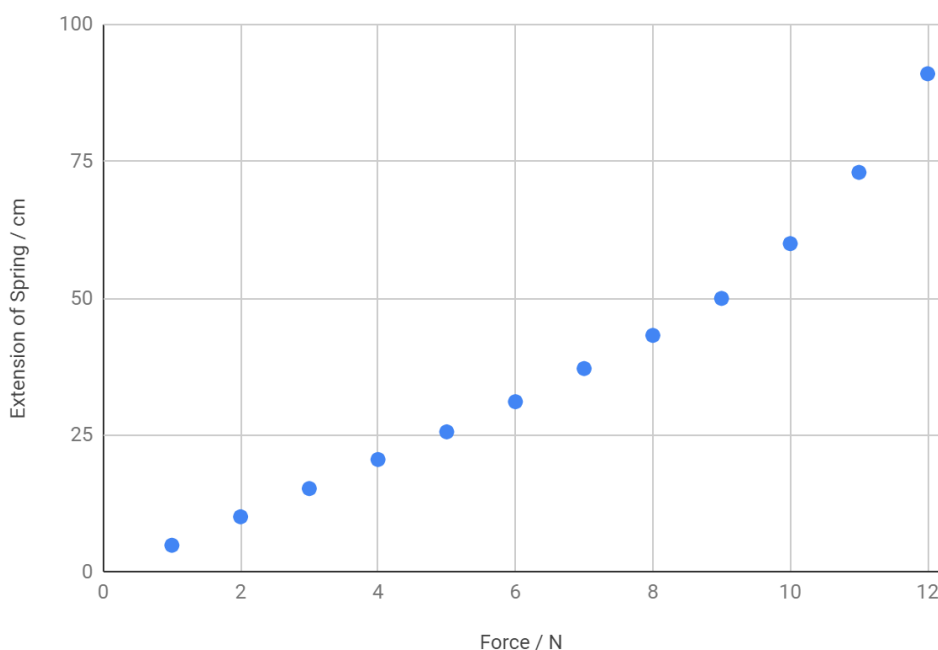
What are the key features of a good results table?

1. Headings should be **underlined**.....
2. Units must be present and are in the **headings**..... of the table
3. Readings are **recorded**..... and an **average**..... is calculated
4. All measurements are written to the same number of **decimal**..... places

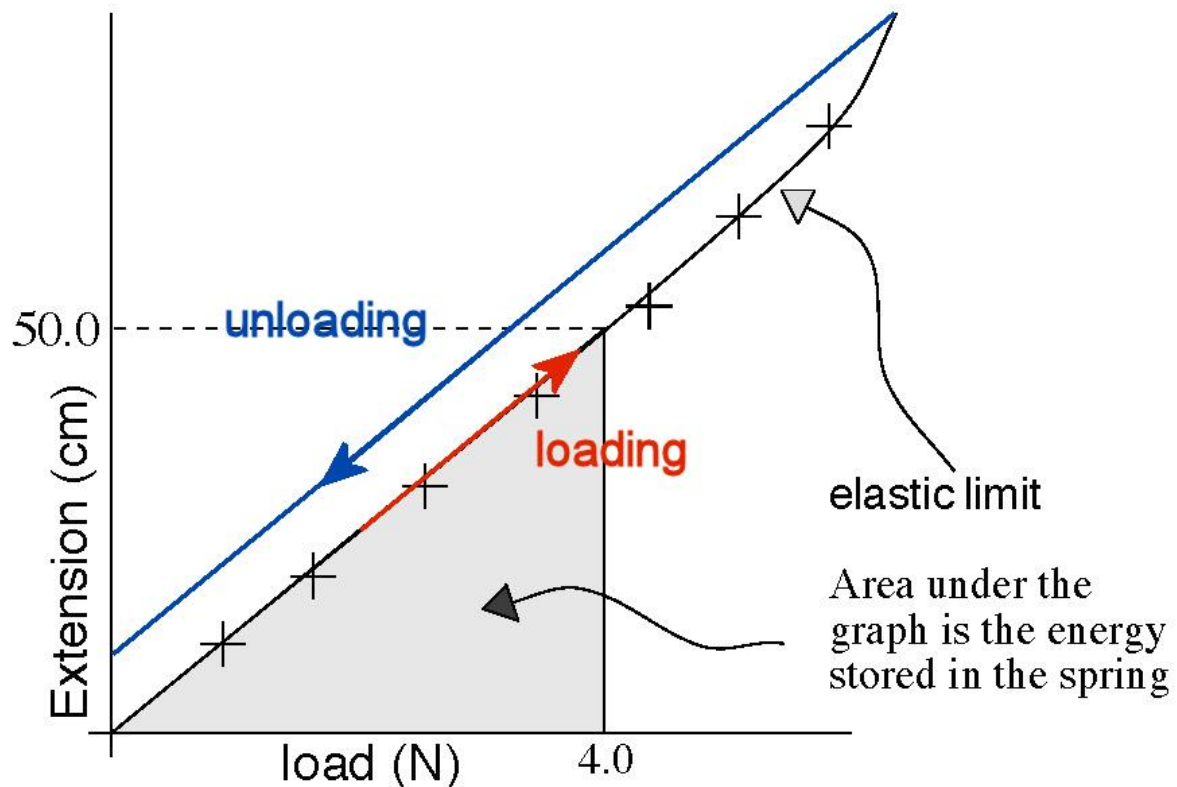
Exemplar Graph

Draw in a line of best fit. Do we use a ruler? Do we draw a curve?

Extension of Spring vs Force



Use a ruler: the line of best fit is straight and passes through the origin for the Hookean region, with roughly equal points either side. Beyond the elastic limit the points curve, so the line curves.



Let's just consider the loading part of the graph:

Conclusion

The **extension** is directly..... proportional..... to the **Force** because the line is a straight..... line that passes through the origin....., but only up to a point which is called the elastic..... limit.

This is HOOKE'S LAW.

Any object that obeys Hooke's Law is said to be Hookian..... .

Hence we know an object is Hookian..... because its Force/Extension graph will be a straight..... line that passes through the origin.....!

Springs and Metal Wires are Hookian.

Mathematical Analysis

In a direct proportionality, doubling the force will double the extension.

Force is directly proportional to extension

Force = spring constant x extension

$$F = k x$$

Note:

- This only applies inside the elastic limit. Beyond the elastic limit the spring is permanently deformed. This is plastic behaviour.
- k is the spring constant which is the **stiffness**..... of the spring.

Definitions:

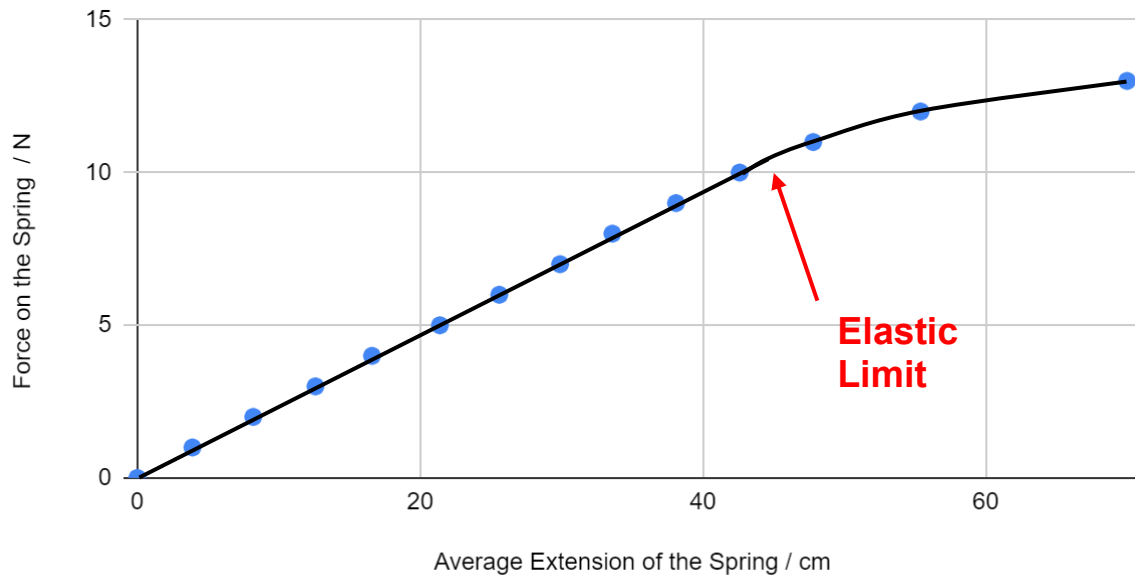
Elastic – can **stretch**..... and will **return**..... to its original shape

Plastic – can **deform**..... and will then **keep**..... the new shape.

Force-Extension Graph

This is often shown this way around:

Force on the Spring / N vs Average Extension of the Spring / cm



Again, the graph is a **straight**..... Line passing through the **origin**..... but only up to the elastic limit.

For the Hookean part of the graph:

$$F = k x$$

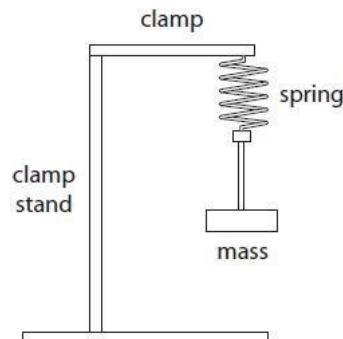
The Spring Constant k will be the gradient of the straight line on the graph above.

Homework option P25 Q1-5

Practice Questions:

A student investigates how the extension of a spring varies when the mass attached to the spring is changed.

(a) The diagram shows most of the equipment the student uses in the investigation.



Describe a method the student could use for the investigation.
Your description should include the measurements taken and how the student could obtain accurate results.

(5)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(b) The student calculates the force each mass applies to the spring.

The table shows the student's results.

(i) Plot the student's results.

(3)

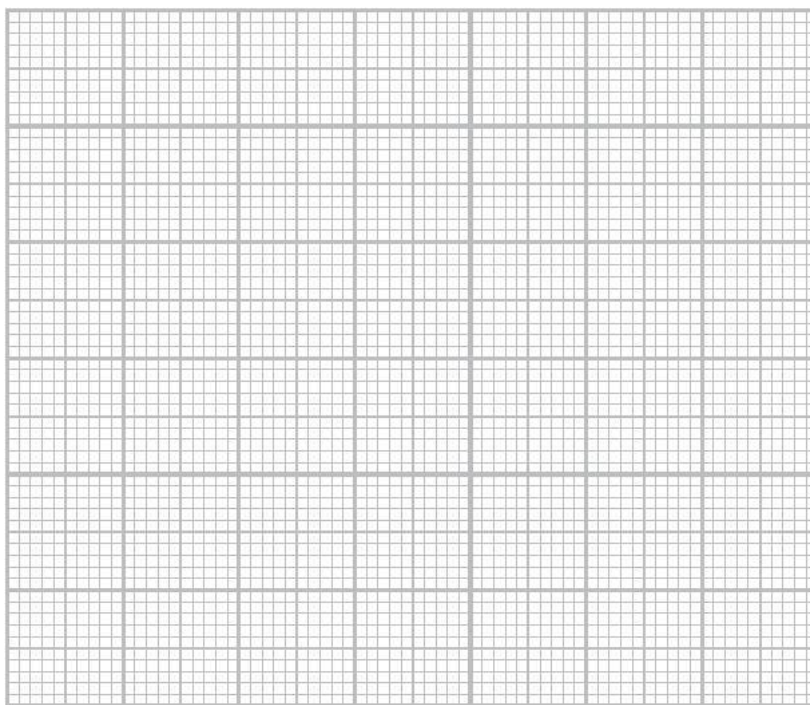
(ii) Draw a circle around the anomalous point.

(1)

(iii) Draw a line of best fit.

(1)

Force in N	Extension in cm
0.0	0.0
1.0	2.5
2.0	5.0
3.0	9.8
4.0	10.0
5.0	12.5
6.0	15.5
7.0	19.5



(iv) Explain whether the spring obeys Hooke's Law.

(3)

.....

.....

.....

.....

.....

.....

(Total for question = 13 marks)

Forces: Lesson 10 : Research Question for Elastic/Rubber Band

We saw that, inside the elastic limit, a metal spring is **Hookian** and **Elastic**. Does this mean that **all** elastic objects are **Hookian**?

Is an elastic band **Hookian**? Does it obey Hooke's Law?

We can test this by producing a Force/extension graph and seeing if it is

a **straight**..... Line through the **origin**.....

No - a rubber band is elastic (it returns to its original shape) but it is not Hookian, because its force-extension graph is a curve, not a straight line through the origin.

No. An elastic band does not obey Hooke's Law: its force-extension graph curves, and the loading and unloading curves do not follow the same path.

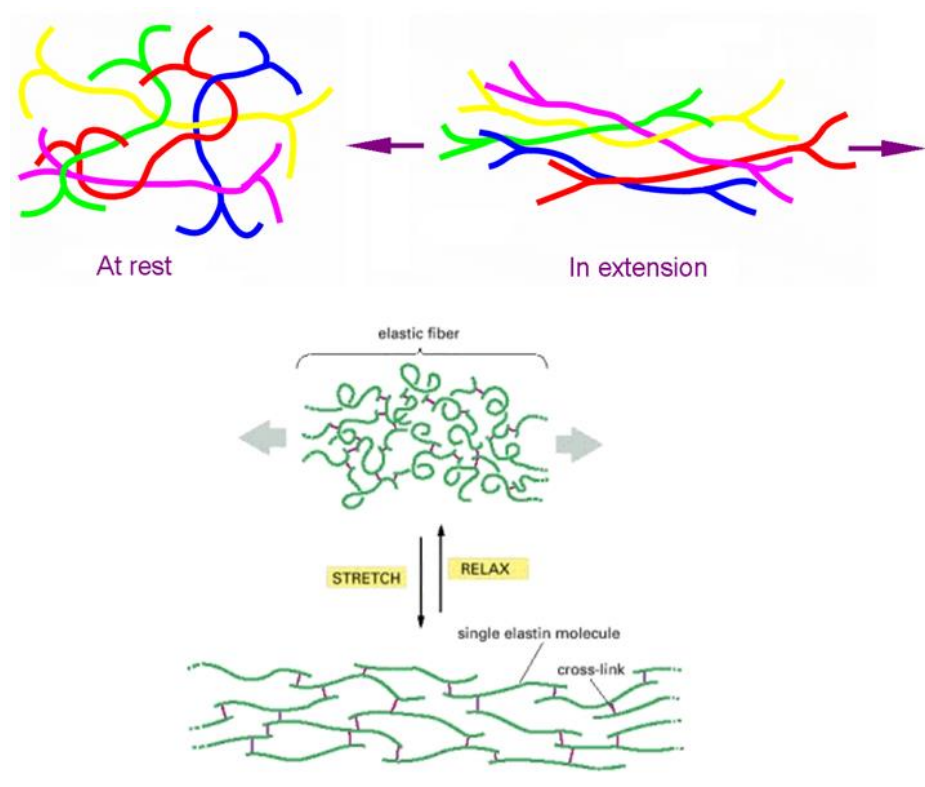
Results

Force /N	Extension /cm (when loading, ie. (adding weight)	Extension /cm (when unloading, ie. (removing weight)
	±	±
0	0	0
1		
2		
3		
4		
5		
6		
7		
8		

Conclusion

An elastic band is **elastic**..... because it can **stretch**..... and **return**..... to its original shape.

The elastic band is **Non-linear**..... because the force/extension graph is not a **straight**..... line, which means that load is not **directly**..... **proportional**..... to extension. The elastic band does not obey Hooke's Law.

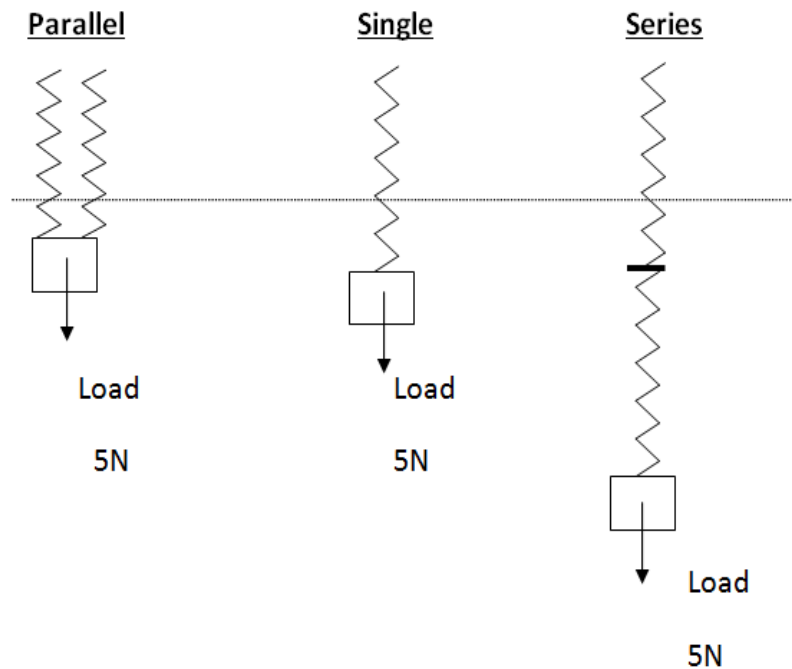


Homework P 74, Q 5 and REVISE

Produce your revision notes (they will be assessed)

Do the Practice Exam Questions on Page 93 onwards in your student notes (answers are on ISLE)

Extension – Springs in Series and Parallel



All 3 systems are given an equal load of 5N.

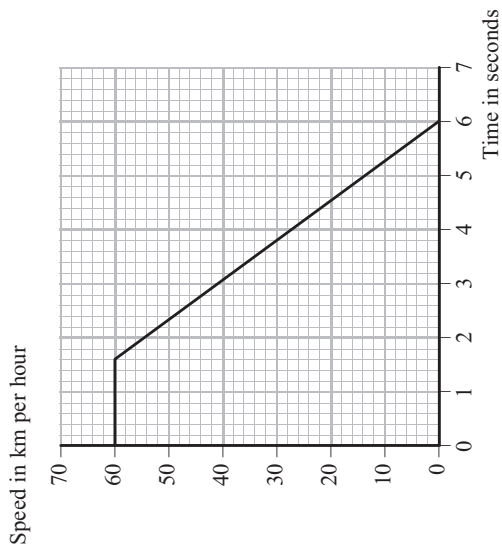
- § Measure the extension for a single spring.
- § What will be the extension of the other 2 systems?
- § What is happening to the stiffness (spring constant)?

Measure the original length and the new length of the single spring with a 5 N load and subtract: extension = new length - original length.

Springs in series: extension = 2 x the extension of a single spring. Springs in parallel: extension = 1/2 the extension of a single spring.

Stiffness (spring constant): the parallel system is twice as stiff ($2k$) and the series system is half as stiff ($k/2$) - so parallel extends less for the same 5 N load.

2. A child runs out in front of a truck. The driver makes an emergency stop. The graph shows the speed of the truck from the time when the driver first sees the child.



- (a) (i) State the time, in seconds, for the driver to react.
Time = 2 s (1)

- (ii) Calculate the time, in seconds, for the brakes to stop the truck.

Time = 6.7 s (1)

- (b) The driver had been drinking alcohol. State the effect this may have had on:

- (i) the time for the driver to react;

..... (1)

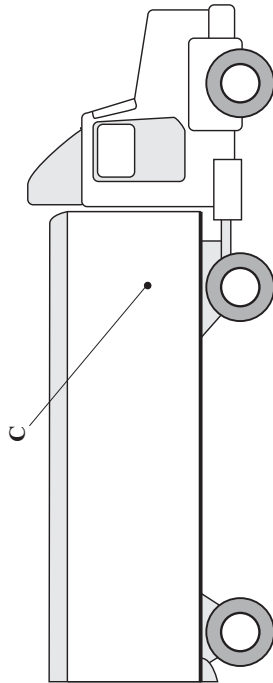
- (ii) the time for the brakes to stop the truck.

..... (1)

- (c) Give one feature of the road conditions which may increase the stopping distance of a vehicle.
icy road (1)

- (d) The diagram shows a truck.

- (i) The centre of gravity of the truck and its load is at the point labelled C. Add to the diagram an arrow showing the total weight. (1)



- (ii) State the name of a force which opposes the motion of the truck.

friction

Q2
(Total 7 marks)

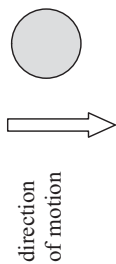
13. In 1907, a tradition started of dropping a ball, made from wood and iron, into Times Square in New York at midnight on New Year's Eve. The ball had a mass of 300 kg and was dropped from a great height.

(a) Calculate the weight, in newtons, of the ball.

$W = m \times g = 300 \times 10 = 3000 \text{ N}$

Weight = 3000 N
(2)

(b) The diagram shows a falling ball.



(i) Add a labelled arrow to the diagram to show the weight of the ball. (1)

(ii) Add another arrow to the diagram to show another force acting on the ball during its fall through the air. (1)

(iii) State the name of this force.

air resistance (1)

The ball falls under gravity, accelerating at about 9.8 m/s^2 (less once air resistance grows); weight = mass $\times$ g.

· Add a labelled arrow starting at the ball and pointing vertically downwards: weight (W).

(c) Eventually the ball reached terminal velocity when it was travelling at constant speed.

Explain why the ball reached terminal velocity.

Weight acts downwards and air resistance acts upwards.

(3)

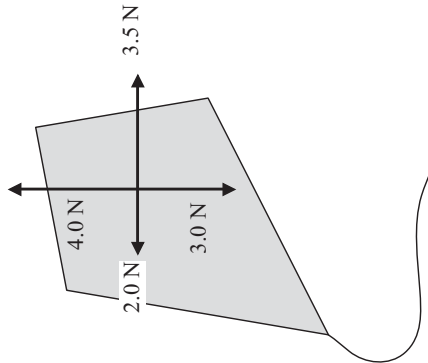
(Total 8 marks)

Q13

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16. The diagram shows a kite in the air and the size and direction of the forces which act on it.



(a) Calculate the resultant upward force, in newtons, on the kite.

.....
Resultant upward force = 0.5 N (1)

(b) Calculate the resultant sideways force, in newtons, on the kite and state its direction.

Resultant sideways force = 2.0 (= 4.0 - 2.0) N
Direction is the direction of the 4.0 N force (2)

Weight acts downwards, tension along the string and air resistance/lift upwards; the kite is in equilibrium when the resultant force is zero.

(c) The air temperature is 17 °C.
Temperature is a scalar quantity.
Explain what is meant by a scalar quantity.

.....
..... (2)

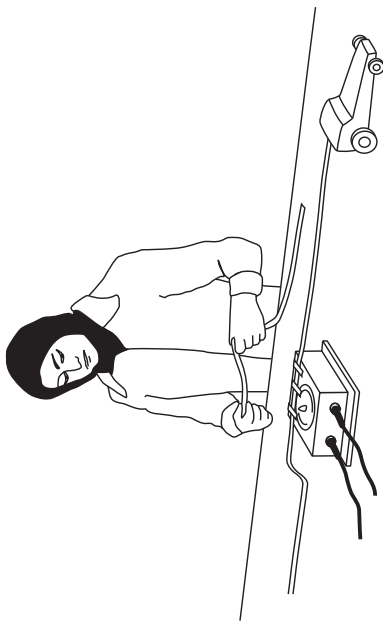
(d) State the word which is used to describe quantities, such as force and acceleration, which are **not** scalar quantities.

vector

(Total 6 marks)

A scalar quantity has magnitude (size) only - it has no direction.

5. A ticker timer is a device which makes dots on a paper tape.
A student fastens one end of the tape to a toy car and uses the ticker timer to record the motion of the toy car.
Her investigation is shown below.



Part of the tape from the student's investigation is shown in the following drawing.



- (a) (i) The ticker timer produces dots at a steady rate of 50 dots per second.

Calculate how long, in seconds, it took the toy car to travel from A to B.

.....
.....

Time taken = 0.2 s
(2)

- (ii) Distance A to B is 73 mm.

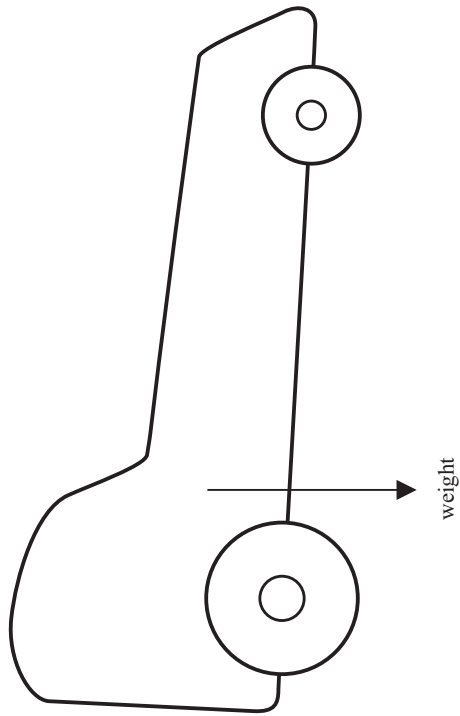
Calculate the average speed, in mm/s, of the car as it travelled this distance.

.....
.....

Average speed = 365 mm/s
(2)

time for 10 dots interval = 0.2 s
time for 1 dot interval = 0.02 s
Distance A to B = 73 mm
Average speed = 365 mm/s
time = 73 / 0.2

- (b) The diagram shows the toy car and the arrow shows where its weight acts.
Put an X on the diagram to show the position of the centre of gravity of the car.
The centre of the X should be at the centre of gravity.



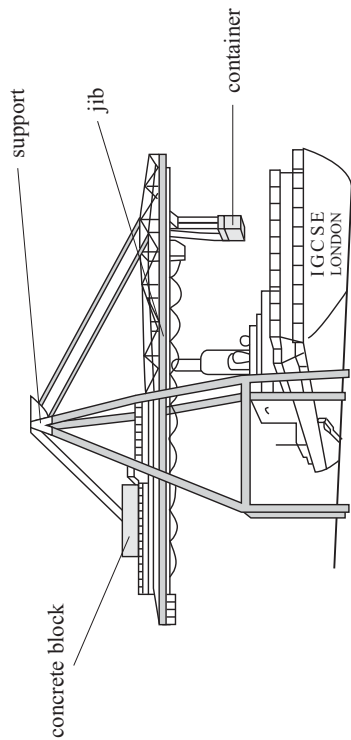
(1) Q5
(Total 5 marks)

Put the X on the line of the weight arrow, in the middle of the car - the weight acts through the centre of gravity.

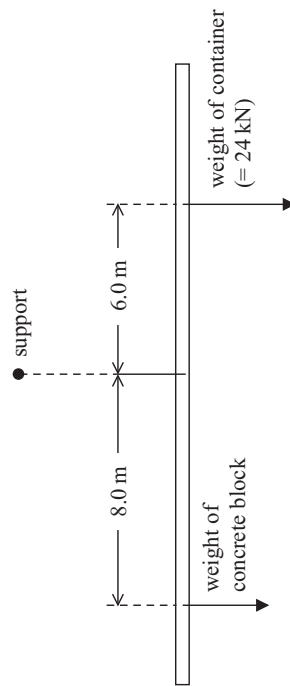
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8. A dockside crane is used to load containers onto a ship.



The jib is designed so that its weight is equally balanced on each side of the support.
The following diagram shows the other turning forces on the jib.
The jib is in equilibrium.



- (a) (i) State the principle of moments for a system which is in equilibrium.

.....
.....
(1)

Leave blank

- (ii) Use the principle of moments to calculate the weight, in kilonewtons, of the concrete block.

.....
.....
.....

Weight of the concrete block = **32** kN (2)

- (b) Another container weighs 26 000 N.

- (i) State the equation which relates weight, mass and the gravitational field strength, g .

.....
(1)

- (ii) Calculate the mass of this container.

Show how you arrived at your answer and state the unit.
 $g = 10 \text{ N/kg}$

.....
.....
.....

Mass of container = **2600** kg (3)

(Total 7 marks)

Q8

moment = force $\times$ perpendicular distance = $26000 \text{ N} \times 0.6 \text{ m} = 15600 \text{ Nm}$
weight = mass $\times$ gravitational field strength = $2600 \text{ kg} \times 10 \text{ N/kg}$

At constant speed the cable tension equals the container's weight ($W = m \times g$); while accelerating upwards, tension = $W + m \times a$.

Page 2 of 2

(Source : November 2010)

9. The photograph shows two people pushing a car. They have to push the car with a large force before it just starts to move.



(a) Name the type of force that opposes motion. (1)

(b) When the car is moving the unbalanced force acting on the car is 150 N.

(i) State the equation which relates unbalanced force, mass and acceleration. (1)

(ii) What does the term **unbalanced force** mean? (1)

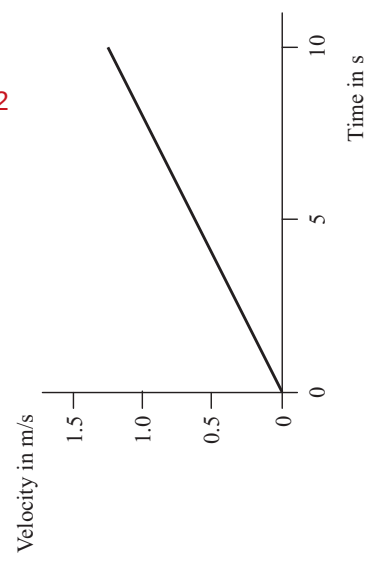
Yes. The graph is a straight line through the origin, so the acceleration is constant, and its gradient = $1.25 \text{ m/s} / 10 \text{ s} = 0.125 \text{ m/s}^2$ - the same value calculated in (b)(iii).

- (iii) The mass of the car is 1200 kg.

Calculate the acceleration of the car and state its unit.

.....
.....
Acceleration (2)
 $a = F / m = 0.125$

- (c) The velocity-time graph shows how the car moves from rest during ten seconds.



Explain whether the graph supports your answer in (b)(iii).

.....
.....
..... (2)

(Total 7 marks)

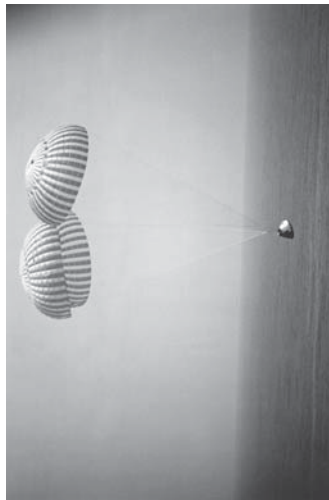
Q9

Leave blank

Forces in the same direction add (e.g. $300 \text{ N} + 300 \text{ N} = 600 \text{ N}$ resultant); $F = m \times a$, so a larger resultant force gives a greater acceleration.

Page 8 of 9

- 15** (a) Parachutes are used to slow down a spacecraft as it falls through the atmosphere.
Photograph **G** shows an Apollo spacecraft with three parachutes attached.



Photograph **G**

This spacecraft falls at a constant velocity.

- (i) State the name of this constant velocity.

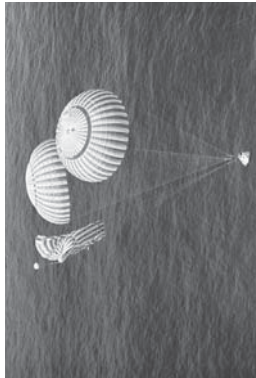
(1)

- (ii) Explain why this velocity stays at a constant value.

(3)

terminal velocity

- (iii) Photograph **H** shows an identical Apollo spacecraft. Only two of its parachutes are working.



Photograph **H**

Explain how the constant velocity reached by this spacecraft compares with the constant velocity of the spacecraft shown in photograph **G**.

(2)

- (b) Photograph **I** shows a Space Shuttle using a parachute when it lands on a runway.



Photograph **I**

Explain what would happen to the stopping distance of the Shuttle if this parachute did not open.

(2)

The stopping distance would be greater

With only two parachutes there is less upward drag, so the spacecraft must fall faster for drag to equal weight - its constant velocity is

greater than that of the spacecraft in photograph **G**.

(Total for Question 15 = 8 marks)

16 The diagram shows the driving force on a sports car as it moves along a race track.



(a) Name **two** forces that oppose the driving force.

(2)

(b) The car has a mass of 1400 kg.

The acceleration of the car is 5.5 m/s^2 .

(i) State the equation linking force, mass and acceleration.

(1)

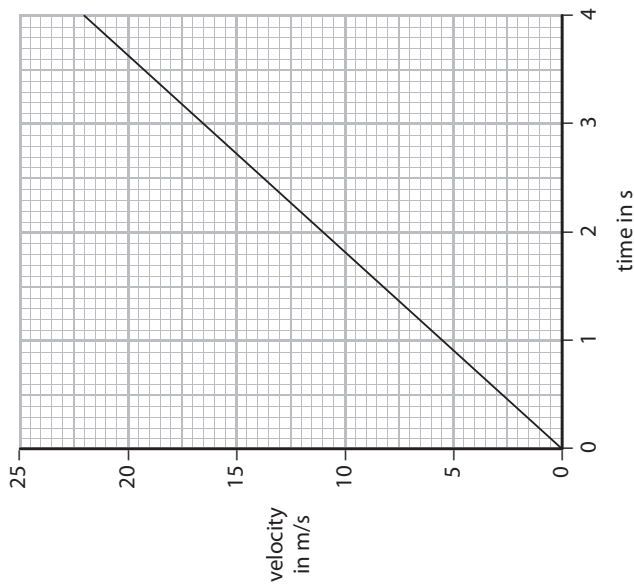
(ii) Calculate the force causing this acceleration.

(2)

Force = N

1700 ($F = m a = 1400 \times 5.5$)

(c) Graph **1** shows how the velocity of the car changes with time.



Graph **1**

Calculate the distance that the car travels in the first four seconds.

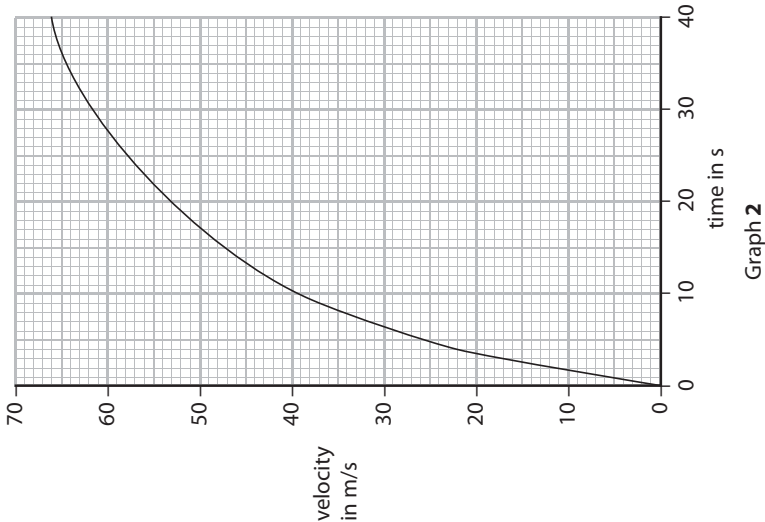
(3)

Distance = m

40 (distance = area under graph = $1/2 \times 4 \times 20$)

QUESTION 16 CONTINUES ON THE NEXT PAGE

(d) As the car travels further along the track, its acceleration changes as shown in graph 2.



(i) Which feature of graph 2 shows that the acceleration changes?

(1)

(ii) The acceleration changes even though the driving force does **not** change.

Suggest **two** possible reasons for this change in acceleration.

(2)

8 Missing force = 3.0 N ($W = m \times g = 0.300 \text{ kg} \times 10 \text{ N/kg}$)

A student investigates the extension of a rubber band when masses are added.
(a) Tick the boxes to select the correct items of apparatus that the student would need in order to complete this investigation.

Two items have already been selected.

(2)

Item	Tick (✓) if item needed
ammeter	
steel spring	
retort stand and clamp	
rubber band	✓
ruler	
thermometer	
mass hanger	
masses	✓

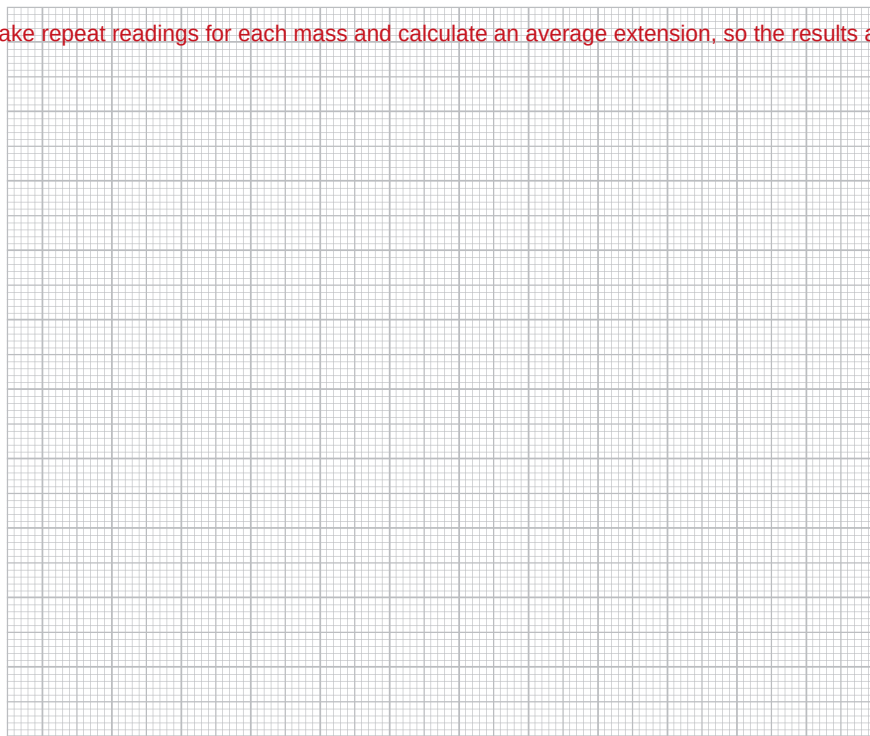
(b) The table below shows the student's results.

Mass in g	Force in N	Extension in cm
0	0	0.0
150	1.5	2.4
350	3.5	6.3
550		12.8
750	7.5	18.6
1050	10.5	24.0

(i) Complete the table by inserting the missing force.

(1)

(ii) Plot a graph to show how force varies with extension.



(5) Take repeat readings for each mass and calculate an average extension, so the results are reliable.

(iii) Use the information from the graph to explain whether the rubber band obeys Hooke's Law.

(2)

5 A student investigates the motion of different falling masses by measuring the time taken for empty cupcake cases to fall from a window.



(a) The student drops one case from the window.

He repeats the experiment with two cases stuck together, then with three cases and then with four.

Name two measuring instruments that he would need for his investigation.

(2)

stopwatch rule

What are the dependent and independent variables in this investigation?

(2)

dependent variable.

independent variable

(c) State one factor that the student should keep constant in order to make this investigation valid (a fair test).

(1)

(d) The student draws this table to record his results.

Add suitable headings to his table.

(2)

time taken in seconds	number of cases in the stack
--	---

(e) State one way that the student can improve his investigation.

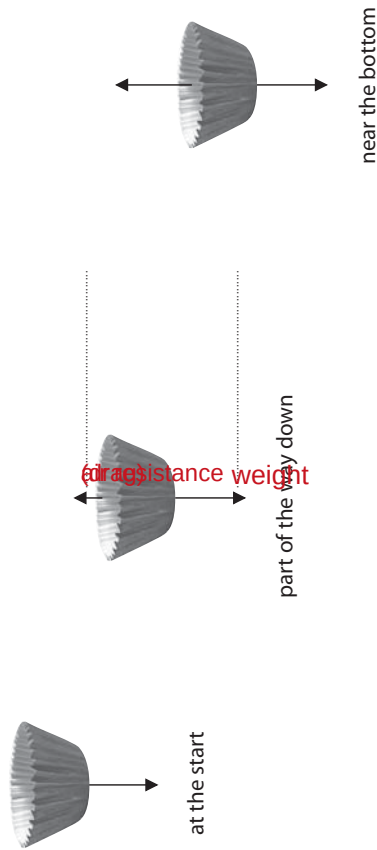
(1)

(f) The student notices that the cases accelerate and then fall at constant speed.

- (i) The arrows in the diagrams show the size and direction of the forces acting on a case at different points in its fall.

Label the forces on the middle diagram.

(2)



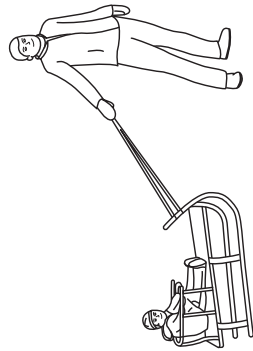
(ii) Explain why the case accelerates and then falls at constant speed.

(3)

Repeat each measurement and calculate an average time - this makes the results more reliable.

Use an electronic timer (or light gate) instead of a stopwatch to reduce the reaction-time error.

7 The diagram shows a man pulling a child on a sledge.



- (a) The acceleration of the sledge is 1.5 m/s^2 .

The mass of the child and sledge is 38 kg.

(i) State the equation linking force, mass and acceleration.

(1)

(ii) Calculate the force needed to produce this acceleration.

(2)

force = N

- (iii) Suggest a reason why the force exerted on the sledge by the man must be greater than the force calculated.

(1)

$F = m \times a = 38 \times 1.5$

Friction between the child's sledge and the ground

(Total for Question 5) 13 marks

(b) The sledge starts from rest and accelerates at 1.5 m/s^2 until its velocity is 2.8 m/s .

(i) State the relationship between acceleration, velocity and time.

(1)

(ii) Show that the time taken to reach 2.8 m/s is about 2 s .

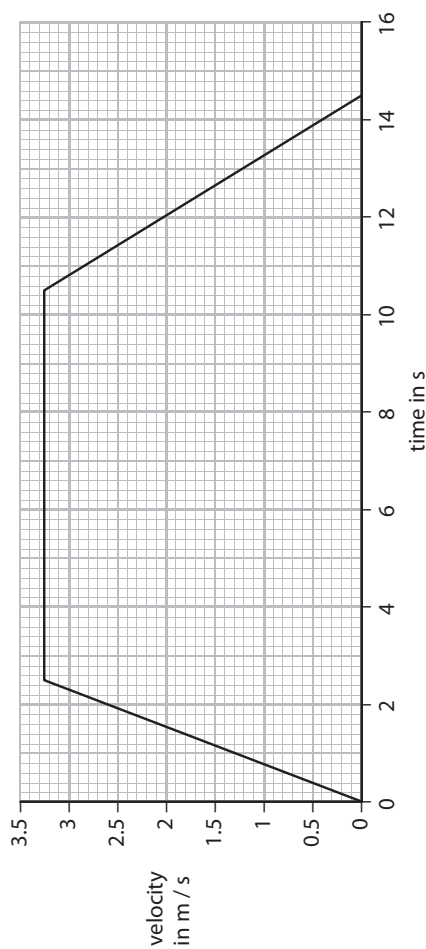
(2)

$$t = v / a = 2.8 / 1.5 = 1.87 \text{ s, which is about } 2 \text{ s.}$$

$$\text{acceleration} = \text{change in velocity} / \text{time taken } (a = (v - u) / t)$$

$$\text{average speed} = \text{distance moved} / \text{time taken}$$

(c) This velocity-time graph shows the motion of the sledge as it travels down a hill.



(i) Calculate the distance travelled by the sledge.

(3)

distance travelled = m

(ii) State the equation linking average speed, distance moved and time taken.

(1)

$$42 \text{ (area under graph} = 1/2 \times 2.8 \times 2.8 + 14 \times 2.8)$$

(iii) Calculate the average speed of the sledge for the whole journey.

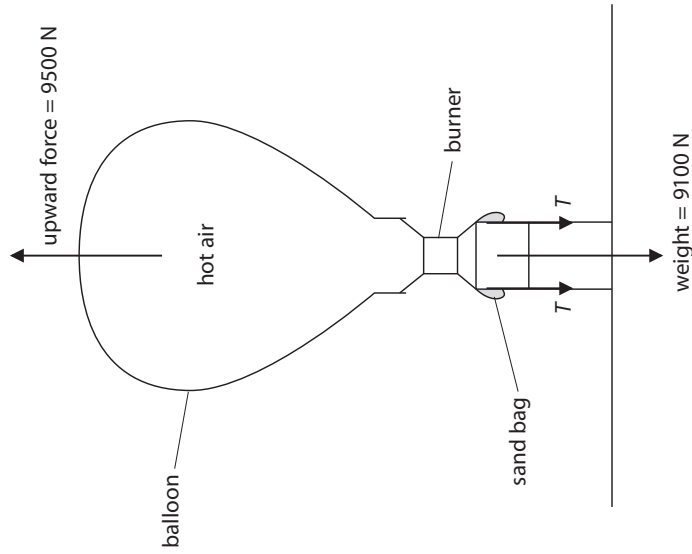
(2)

average speed = m/s

2.6 (2 / 16)

(Total for Question 7 = 13 marks)

7.1 A hot-air balloon is tied to the ground by two ropes.
The diagram shows the forces acting on the balloon.
The tension T in each rope is 200 N.



The ropes are untied and the balloon starts to move upwards.

(a) State the value of the force acting downwards on the balloon immediately after the ropes are untied and before the balloon starts moving.

(1)

force downwards = N

(b) (i) State the relationship between unbalanced force, mass and acceleration.

(1)

unbalanced force = mass x acceleration ($F = m \times a$)

- (ii) The balloon has a total mass of 910 kg.

The initial unbalanced force on the balloon is 400 N upwards.

Calculate the initial acceleration.

(2)

initial acceleration = m/s²

0.44

- (c) Explain how the upward acceleration of the balloon changes during the first few seconds of its flight.

(3)

- (d) While the balloon is still accelerating, the pilot controls the balloon by pouring some sand from the bags.

Explain how this affects the upward acceleration of the balloon.

(2)

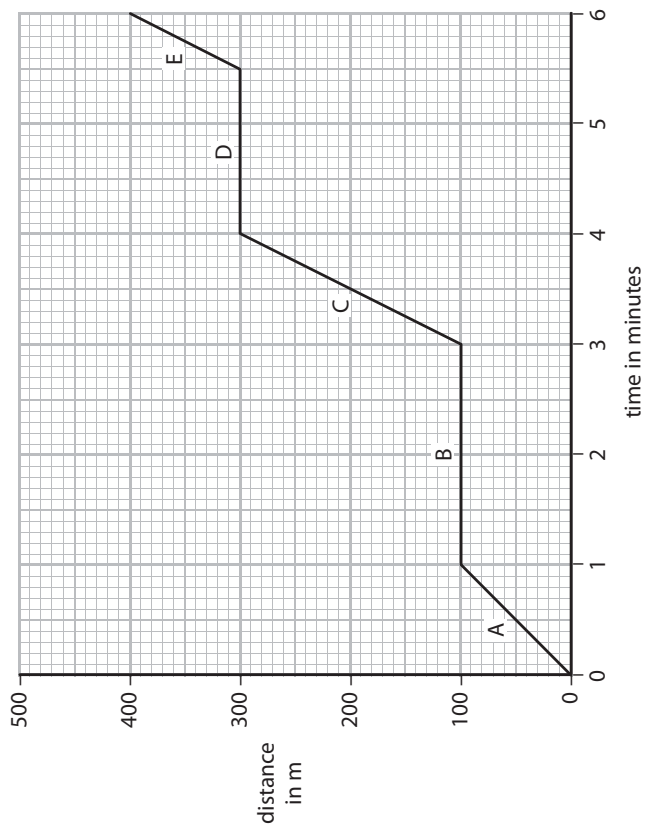
(Total for Question 11 = 9 marks)

- 2 A car travels along a very busy road.



© AnRo0002 (Wikipedia)

The graph shows how the distance travelled by the car changes during a six-minute period.



- (a) Calculate the total amount of time the car is stationary during this period.

(2)

time = 1.0 minutes

(Source : May 2017 - Paper 1PR)

(b) Explain which stage of the graph, A, B, C, D or E, shows the car moving at the slowest speed. (2)

.....

.....

.....

(c) (i) State the equation linking average speed, distance moved and time taken. (1)

(ii) Calculate the speed of the car at stage C.
Give a suitable unit for your answer. (3)

speed = 1.7 unit m/s
State two factors that could affect the braking distance of the car. (2)

.....

.....

(Total for Question 2 = 10 marks)

Stage B. B has the shallowest slope (smallest gradient) of the distance-time graph, so the car covers the least distance each minute - the slowest speed.

average speed = distance moved / time taken

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

- p31: An object accelerates if there is an UNBALANCED (resultant) force on it - the force changes its speed, its direction, or both. Objects with balanced forces (e.g. equal and opposite) carry on at constant speed/direction and do NOT accelerate.
- p66: Same spring, same mass values, same ruler/measuring technique, same starting length of spring, same temperature, and the same wait for the spring to settle each time.
- p72: Hang masses on the spring. Measure the length with a ruler at eye level (avoid parallax); extension = new length - original length. Add masses one at a time, repeat each reading 3 times and take a mean.
- p72: The anomalous result is the one that lies furthest from the pattern of the other points - circle that single point.
- p72: Draw one straight line of best fit through the origin with equal numbers of points above and below it, ignoring the anomalous point.
- p73: Up to the elastic limit the graph is a straight line through the origin, so force is proportional to extension and the spring obeys Hooke's Law. Beyond that point the graph curves, so it no longer obeys the law.
- p79: Add a labelled arrow pointing vertically upwards (opposite to the direction of motion): drag / air resistance.
- p82: For a system in equilibrium, the sum of the clockwise moments about a point equals the sum of the anticlockwise moments about the same point.
- p83: The unbalanced force is the resultant force: the single force left when all the forces acting on the object are added together, taking their directions into account.
- p84: The weight acts downwards and the upward air resistance (drag) from the parachutes increases as it falls. The drag becomes equal to the weight, so the resultant force is zero, the acceleration is zero and the velocity stays constant.
- p87: For small loads the graph is a straight line through the origin, so extension is proportional to force and the rubber band obeys Hooke's Law. For larger loads the graph curves (extension is no longer proportional to force), so it does not obey Hooke's Law.
- p87: Measure the original length of the rubber band, hang each mass on it, measure the new length with the ruler and record the extension (new length - original length).
- p91: The balloon's total mass decreases, so its weight decreases. The upward (lifting) force is now bigger than the weight, so the resultant upward force increases and the acceleration increases.